

1

- (i) Express $x^2 - 5x + 6$ in the form $(x - a)^2 - b$. Hence state the coordinates of the turning point of the curve $y = x^2 - 5x + 6$.

[4]

- (ii) Find the coordinates of the intersections of the curve $y = x^2 - 5x + 6$ with the axes and sketch this curve.

[4]

- (iii) Solve the simultaneous equations $y = x^2 - 5x + 6$ and $x + y = 2$. Hence show that the line $x + y = 2$ is a tangent to the curve $y = x^2 - 5x + 6$ at one of the points where the curve intersects the axes.

[4]

2

- (i) Points A and B have coordinates $(-2, 1)$ and $(3, 4)$ respectively. Find the equation of the perpendicular bisector of AB and show that it may be written as $5x + 3y = 10$.

[6]

- (ii) Points C and D have coordinates $(-5, 4)$ and $(3, 6)$ respectively. The line through C and D has equation $4y = x + 21$. The point E is the intersection of CD and the perpendicular bisector of AB. Find the coordinates of point E.

[3]

- (iii) Find the equation of the circle with centre E which passes through A and B. Show also that CD is a diameter of this circle.

[5]



3 The circle $(x - 3)^2 + (y - 2)^2 = 20$ has centre C.

(i) Write down the radius of the circle and the coordinates of C.

[2]

(ii) Find the coordinates of the intersections of the circle with the x- and y-axes.

[5]

(iii) Show that the points A(1,6) and B(7,4) lie on the circle. Find the coordinates of the midpoint of AB. Find also the distance of the chord AB from the centre of the circle.

[5]

4 Find the coordinates of the point of intersection of the lines $y = 3x - 2$ and $x + 3y = 1$.

[4]

5 Find the equation of the line which is perpendicular to the line $y = 2x - 5$ and which passes through the point (4, 1). Give your answer in the form $y = ax + b$.

[3]

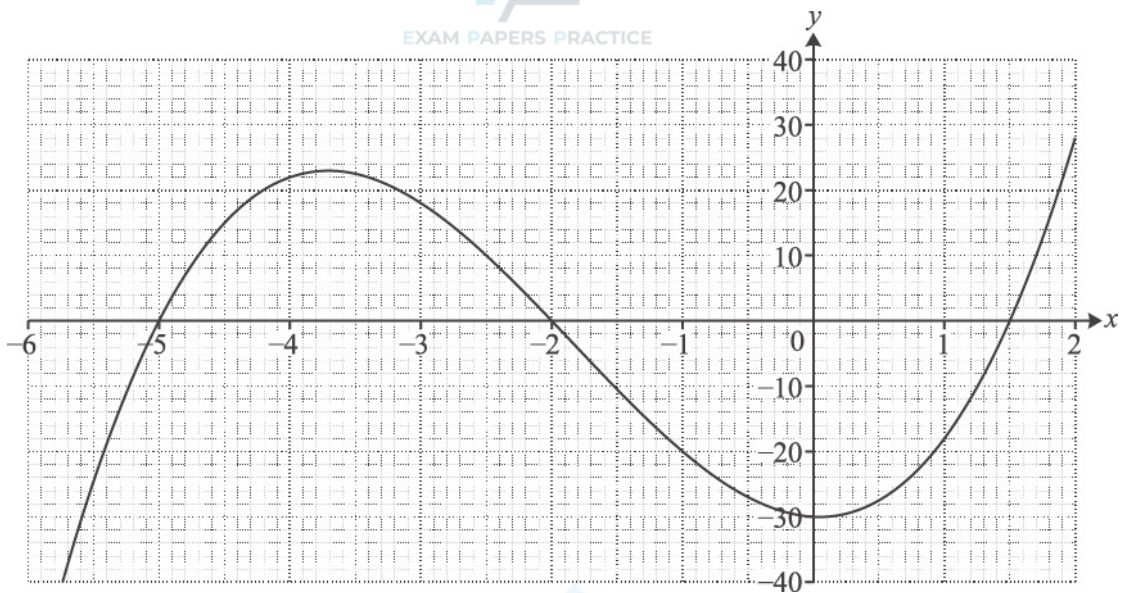


Fig. 12

Fig. 12 shows the graph of a cubic curve. It intersects the axes at $(-5, 0)$, $(-2, 0)$, $(1.5, 0)$ and $(0, -30)$.

- (i) Use the intersections with both axes to express the equation of the curve in a factorised form.

[2]

- (ii) Hence show that the equation of the curve may be written as $y = 2x^2 + 11x^2 - x - 30$.

[2]

- (iii) Draw the line $y = 5x + 10$ accurately on the graph. The curve and this line intersect at $(-2, 0)$; find graphically the x -coordinates of the other points of intersection.

[3]

- (iv) Show algebraically that the x -coordinates of the other points of intersection satisfy the equation

$$2x^2 + 7x - 20 = 0.$$

Hence find the exact values of the x -coordinates of the other points of intersection.

[5]

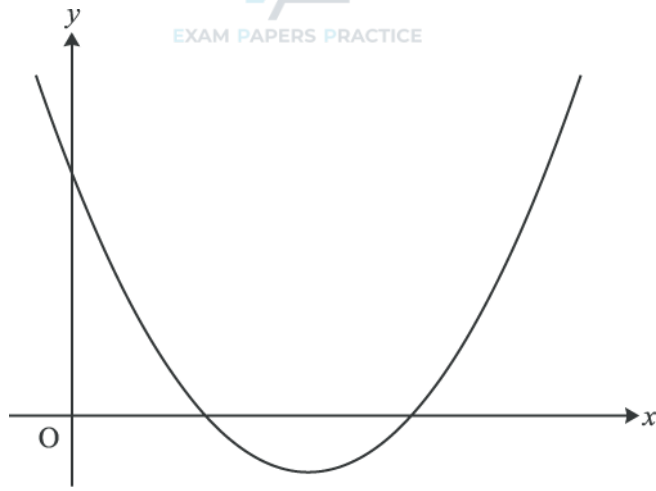
**Fig. 11**

Fig. 11 shows a sketch of the curve with equation $y = (x - 4)^2 - 3$.

- (i) Write down the equation of the line of symmetry of the curve and the coordinates of the minimum point. [2]
- (ii) Find the coordinates of the points of intersection of the curve with the x -axis and the y -axis, using surds where necessary. [4]
- (iii) The curve is translated by $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Show that the equation of the translated curve may be written as $y = x^2 - 12x + 33$. [2]
- (iv) Show that the line $y = 8 - 2x$ meets the curve $y = x^2 - 12x + 33$ at just one point, and find the coordinates of this point. [5]

- 8 Fig. 10 shows a sketch of a circle with centre C (4, 2). The circle intersects the x-axis at A (1, 0) and at B.

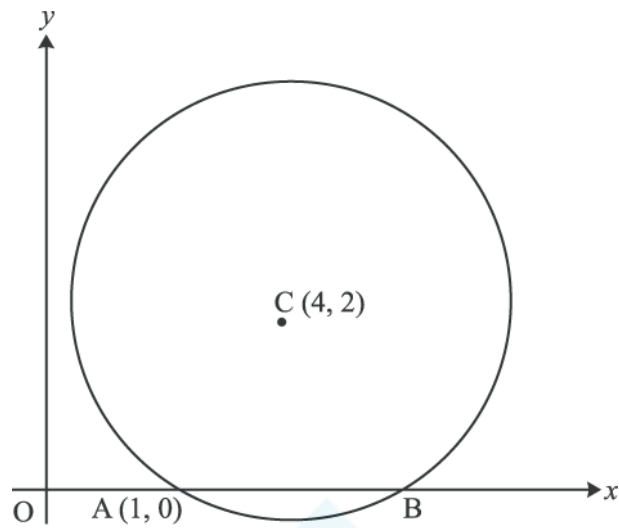


Fig. 10

- (i) Write down the coordinates of B.

[1]

- (ii) Find the radius of the circle and hence write down the equation of the circle.

[4]

- (iii) AD is a diameter of the circle. Find the coordinates of D.

[2]

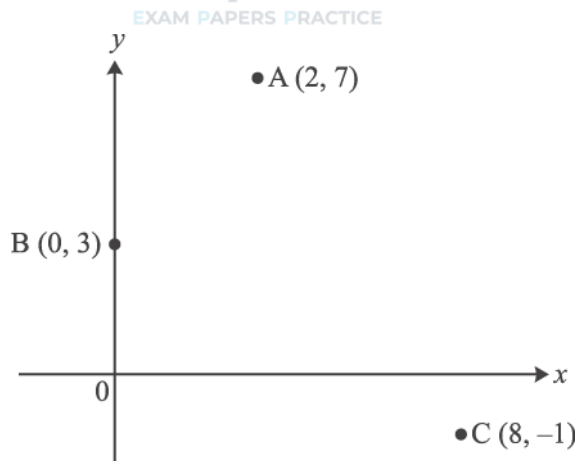
- (iv) Find the equation of the tangent to the circle at D. Give your answer in the form $y = ax + b$.

[4]

- 9 A is the point (1, 5) and B is the point (6, -1). M is the midpoint of AB. Determine whether the line with equation $y = 2x - 5$ passes through M.

[3]

- 10 Fig. 10 shows a sketch of the points A (2, 7), B (0, 3) and C (8, -1).



- (i) Prove that angle ABC is 90° .

[3]

- (ii) Find the equation of the circle which has AC as a diameter.

[4]

- (iii) Find the equation of the tangent to this circle at A. Give your answer in the form $ay = bx + c$, where a , b and c are integers.

[4]

- 11 A and B are points on the curve $y = 4\sqrt{x}$. Point A has coordinates (9, 12) and point B has x -coordinate 9.5. Find the gradient of the chord AB.

The gradient of AB is an approximation to the gradient of the curve at A. State the x -coordinate of a point C on the curve such that the gradient of AC is a closer approximation.

[3]

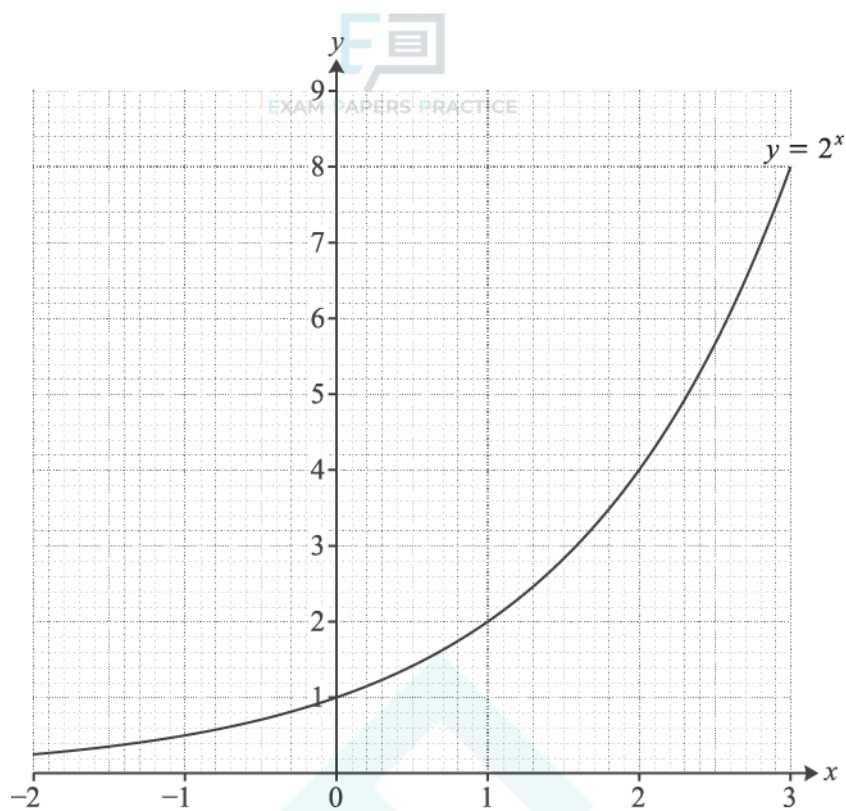
**Fig. 5**

Fig. 5 shows the graph of $y = 2^x$.

- (i) On the copy of Fig. 5, draw by eye a tangent to the curve at the point where $x = 2$. Hence find an estimate of the gradient of $y = 2^x$ when $x = 2$. [3]
- (ii) Calculate the y -values on the curve when $x = 1.8$ and $x = 2.2$. Hence calculate another approximation to the gradient of $y = 2^x$ when $x = 2$. [2]

13

The gradient of a curve is given by $\frac{dy}{dx} = 4x + 3$. The curve passes through the point (2, 9).

- (i) Find the equation of the tangent to the curve at the point (2, 9).

[3]

- (ii) Find the equation of the curve and the coordinates of its points of intersection with the x-axis. Find also the coordinates of the minimum point of this curve.

[7]

- (iii) Find the equation of the curve after it has been stretched parallel to the x-axis with scale factor $\frac{1}{2}$. Write down the coordinates of the minimum point of the transformed curve.

[3]

14

- (i) Calculate the gradient of the chord of the curve $y = x^2 - 2x$ joining the points at which the values of x are 5 and 5.1.

[2]

- (ii) Given that $f(x) = x^2 - 2x$, find and simplify $\frac{f(5+h) - f(5)}{h}$.

[4]

- (iii) Use your result in part (ii) to find the gradient of the curve $y = x^2 - 2x$ at the point where $x = 5$, showing your reasoning.

[2]

- (iv) Find the equation of the tangent to the curve $y = x^2 - 2x$ at the point where $x = 5$.
Find the area of the triangle formed by this tangent and the coordinate axes.

[5]

- 15 In lines 23 and 24 it says that “if the Pythagorean distance between two points A and B is $d(A,B)$ then the taxicab distance satisfies the inequalities $d(A,B) \leq t(A,B) \leq \sqrt{2} \times d(A,B)$.”

This question is about using this result in generalised taxicab geometry.

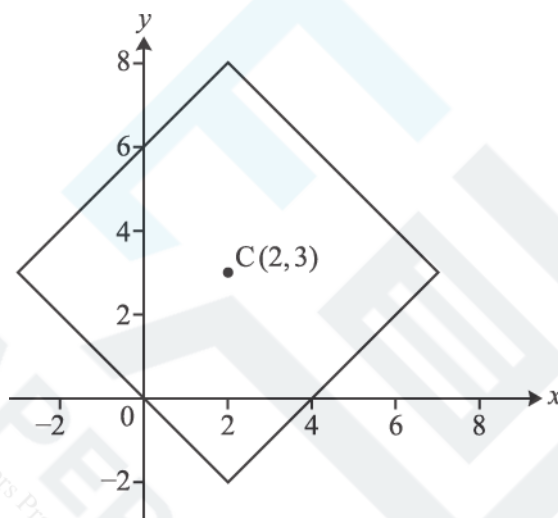
- (i) Given that A is the point (0,0), describe all possible positions of B for which $d(A,B) = t(A,B)$.

[1]

- (ii) Given that A is the point (0,0), describe all possible positions of B for which $t(A,B) = \sqrt{2} \times d(A,B)$.

[2]

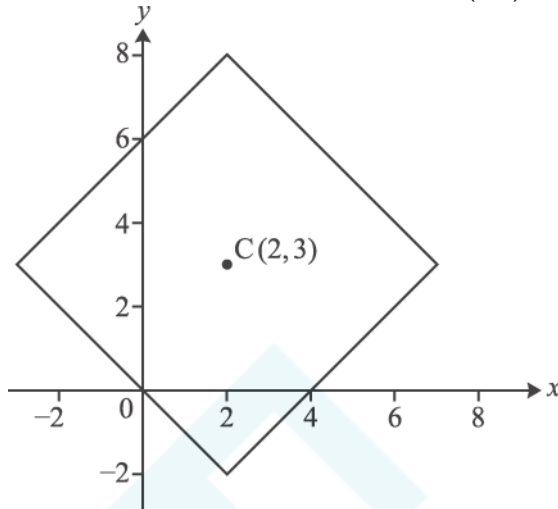
16 Fig. 7 is reproduced below.



- (i) Two points on this locus have x-coordinate -0.7 . Write down the coordinates of each of these points.

- (ii) In lines 77 to 78 it says “adding a second taxicab circle with centre $(2,0)$ and radius 2 shows that in generalised taxicab geometry two different circles can have an infinite number of points in common!”

On the copy of Fig. 7 given below, draw the taxicab circle with centre $(2,0)$ and radius 2.

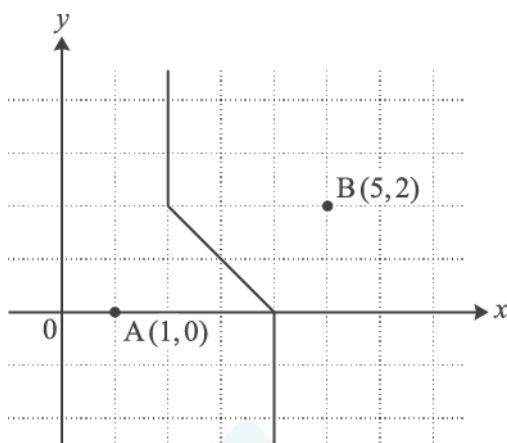


18 Referring to Fig. 5, or otherwise, find the value of $n(4,4)$.

[2]

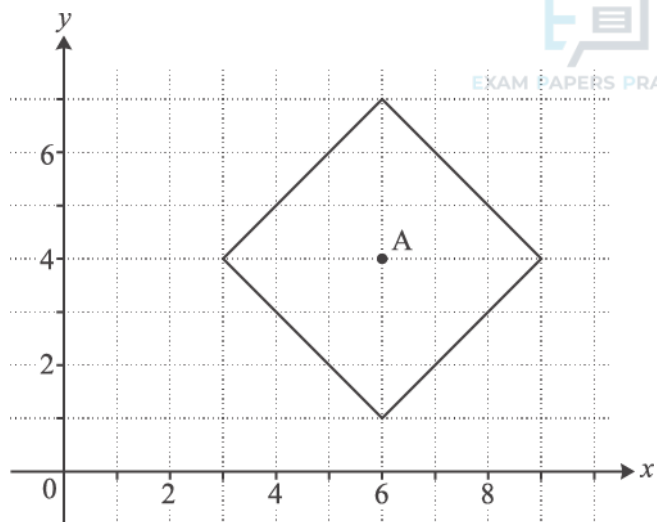
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- (i) Describe the following locus of a point P, using the notation $t(P,A)$ and $t(P,B)$ as appropriate.



[1]

- (ii) Describe the following locus of a point P, using the notation $t(P,A)$ as appropriate.



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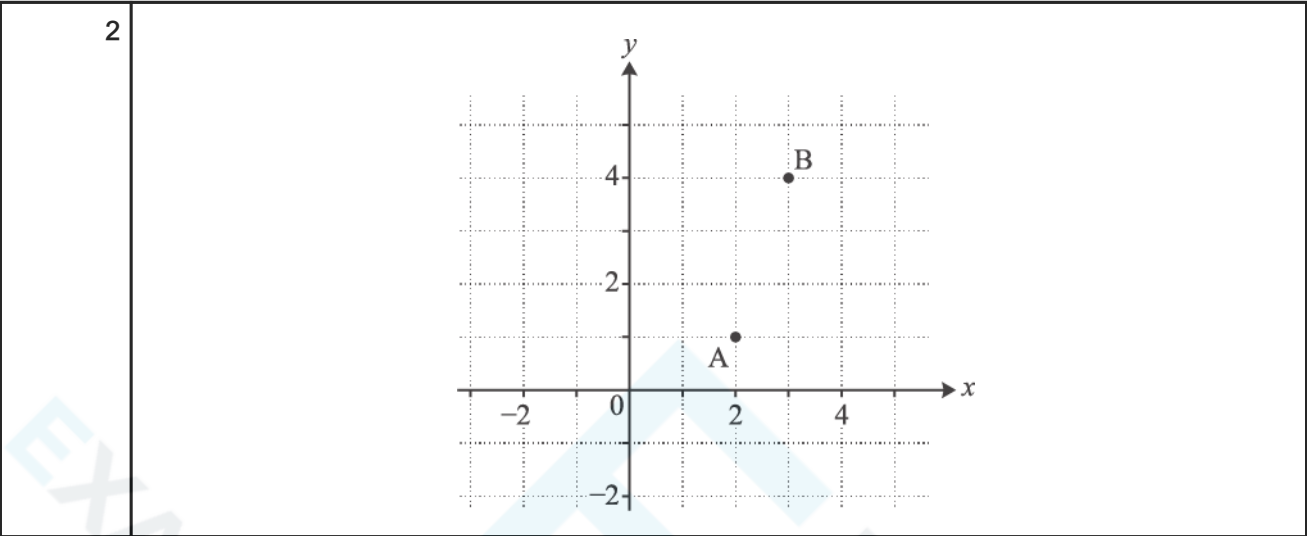
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[2]

20 This question is concerned with generalised taxicab geometry.

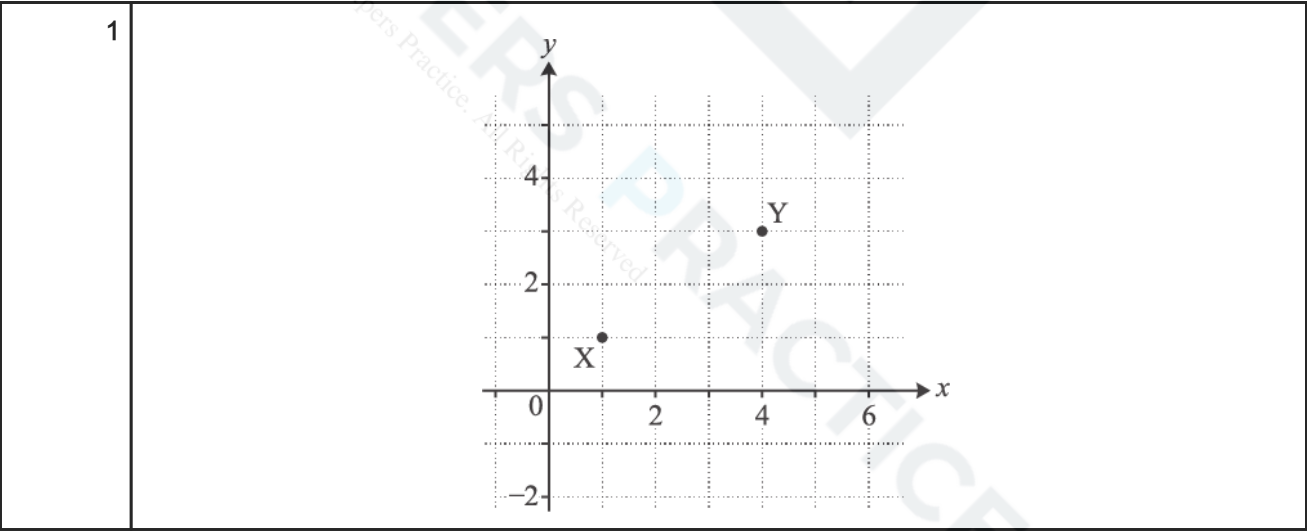
On the grid below, show the locus of a point P where $t(P,A) = t(P,B)$.

[3]

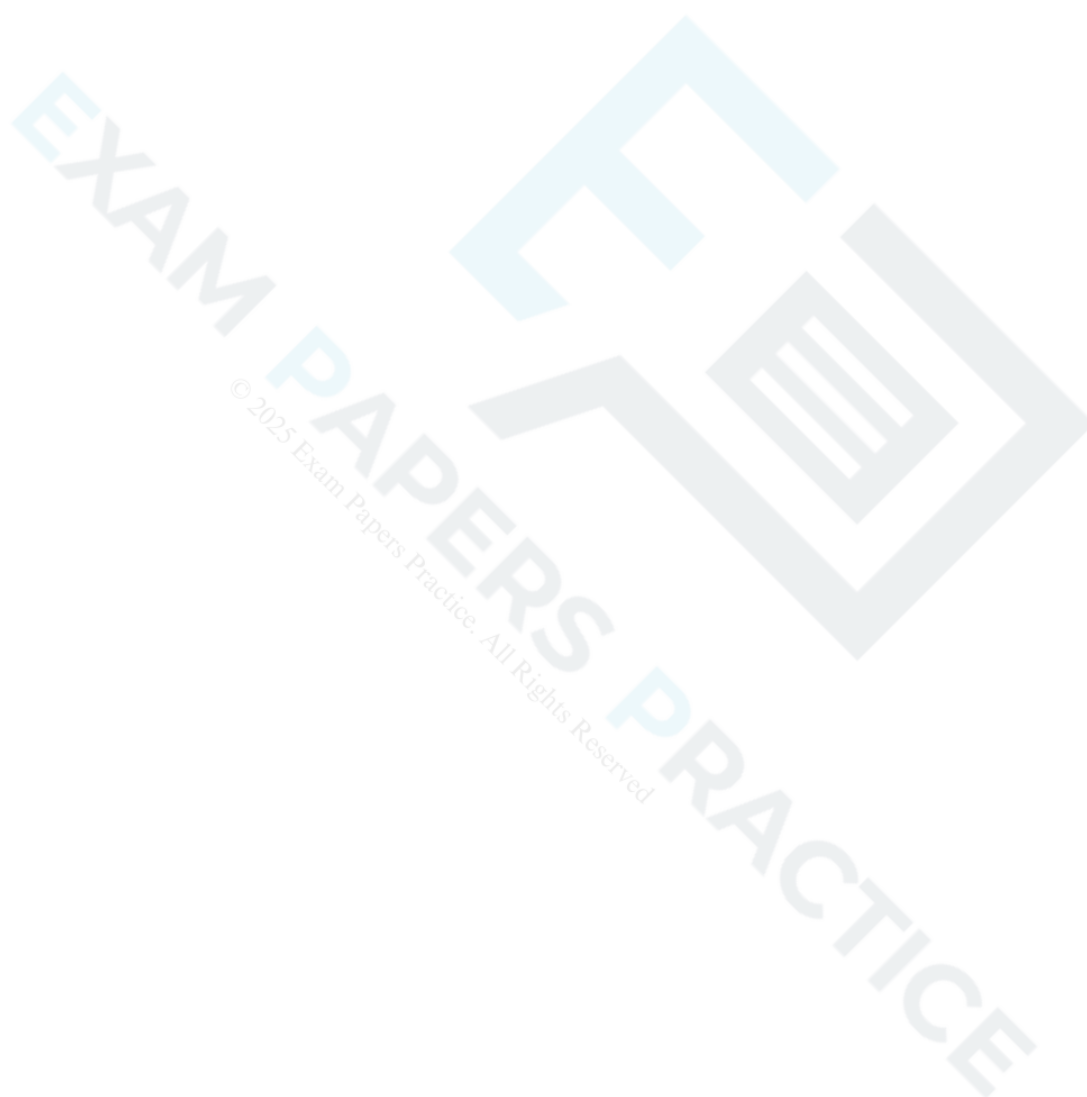


21 On the grid below mark all three possible positions of the point P with integer coordinates for which $t(P,X) = 4$ and $t(P,Y) = 3$.

[3]



- 22 Two points, A and B, have position vectors $\mathbf{a} = \mathbf{i} - 3\mathbf{j}$ and $\mathbf{b} = 4\mathbf{i} + 3\mathbf{j}$. The point C lies on the line $y = 1$. The lengths of the line segments AC and BC are equal. Determine the position vector of C. [4]



23 A circle has equation $(x - 2)^2 + (y + 3)^2 = 25$.



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(a) Write down

- The radius of the circle.
- The coordinates of the centre of the circle.

[2]

(b) Find, in exact form, the coordinates of the points of intersection of the circle with the y -axis.

[3]

(c) Show that the point $(1, 2)$ lies outside the circle.

[2]

(d) The point $P(-1, 1)$ lies on the circle. Find the equation of the tangent to the circle at P .

[4]

- (a) Sketch the graph of $y = \frac{1}{x} + a$, where a is a positive constant.
- State the equations of the horizontal and vertical asymptotes.
 - Give the coordinates of any points where the graph crosses the axes.

[4]

- (b) Find the equation of the normal to the curve $y = \frac{1}{x} + 2$ at the point where $x = 2$.

[5]

- (c) Find the coordinates of the point where this normal meets the curve again.

[3]

25 In this question you must show detailed reasoning.

Determine for what values of k the graphs $y = 2x^2 - kx$ and $y = x^2 - k$ intersect.

[6]

26 In this question you must show detailed reasoning.

Find the coordinates of the points of intersection of the curve $y = x^2 + x$ and the line $2x + y = 4$.

[5]

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27 Fig. 16.1, Fig. 16.2 and Fig. 16.3 show some data about life expectancy, including some from the pre-release data set (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>).

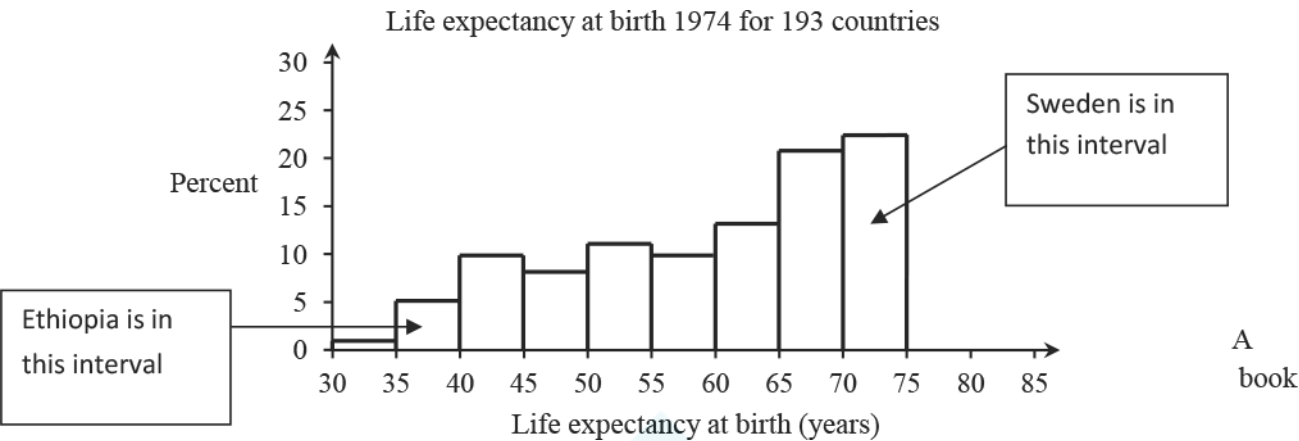


Fig. 16.1

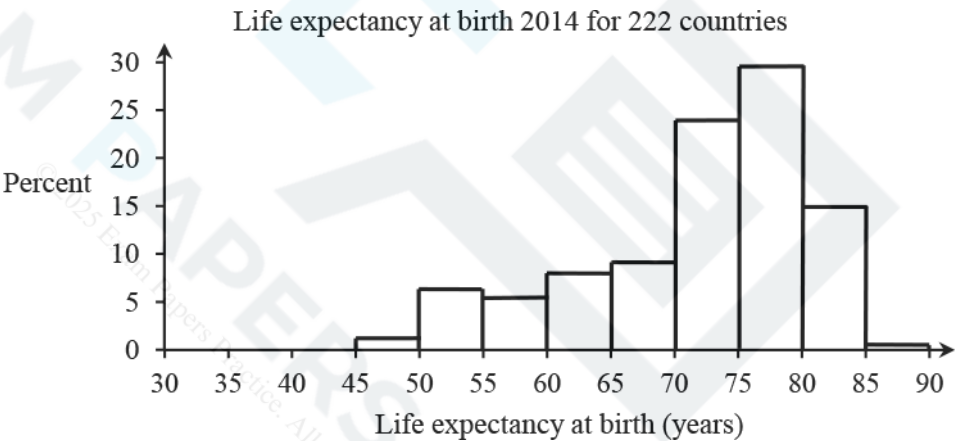


Fig. 16.2



Increase in life expectancy from 1974 to 2014 (years)

Increase in life expectancy for 193 countries from 1974 to 2014	
Number of values	193
Minimum	- 4.618
Lower quartile	6.9576
Median	9.986
Upper quartile	15.873
Maximum	30.742

Source: CIA World Factbook and Gapminder

- (a) Comment on the shapes of the distributions of life expectancy at birth in 2014 and 1974. [2]
- (b) (i) The minimum value shown in the box plot is negative. What does a negative value indicate? [1]
- (ii) What feature of Fig 16.3 suggests that a Normal distribution would **not** be an appropriate model for increase in life expectancy from one year to another year? [1]
- (iii) Software has been used to obtain the values in the table in Fig. 16.3. Decide whether the level of accuracy is appropriate. Justify your answer. [1]
- (iv) John claims that for half the people in the world their life expectancy has improved by 10 years or more. Explain why Fig. 16.3 does not provide conclusive evidence for John's claim. [1]
- (c) Decide whether the maximum increase in life expectancy from 1974 to 2014 is an outlier. Justify your answer. [3]

Here is some further information from the pre-release data set (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>).

Country	Life expectancy at birth in 2014
Ethiopia	60.8
Sweden	81.9

(d) (i) Estimate the change in life expectancy at birth for Ethiopia between 1974 and 2014.

(ii) Estimate the change in life expectancy at birth for Sweden between 1974 and 2014.

(iii) Give **one** possible reason why the answers to parts (i) and (ii) are so different.

[4]

Fig.16.4 shows the relationship between life expectancy at birth in 2014 and 1974.

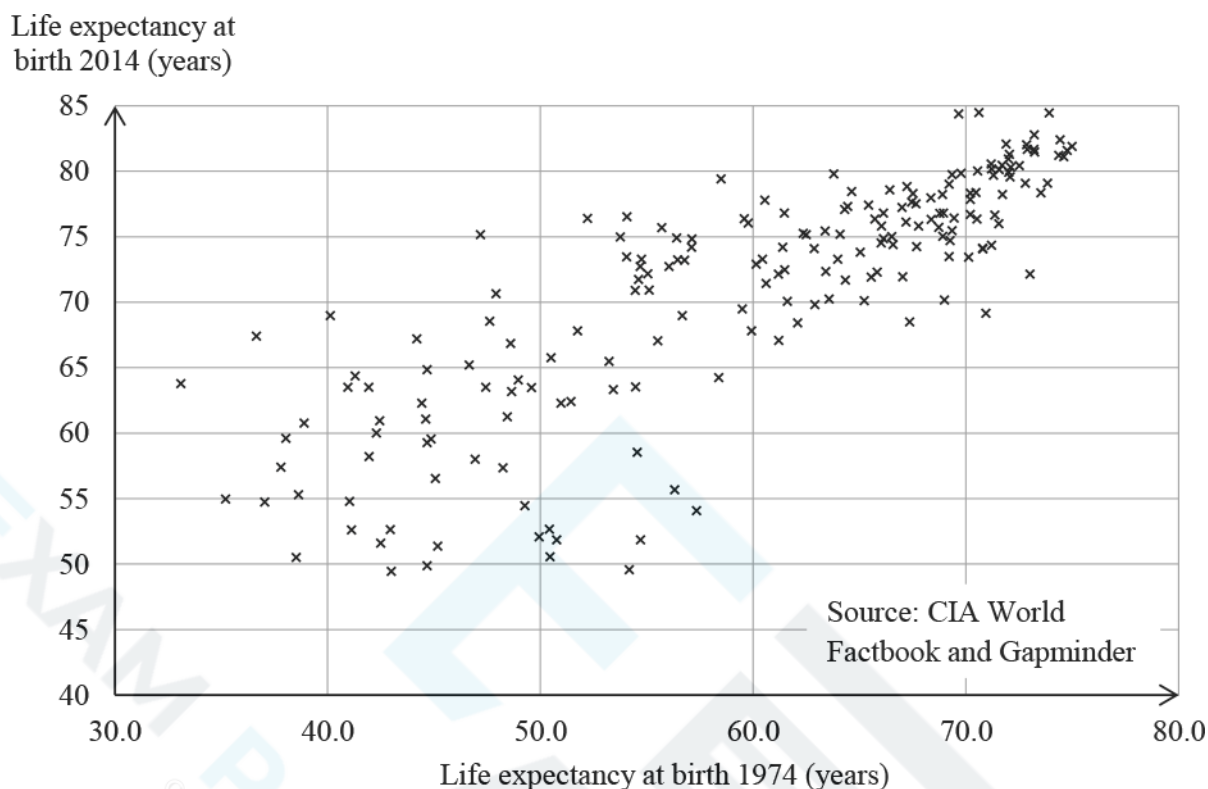


Fig. 16.4

A spreadsheet gives the following linear model for all the data in Fig 16.4.

$$(\text{Life expectancy at birth 2014}) = 30.98 + 0.67 \times (\text{Life expectancy at birth 1974})$$

The life expectancy at birth in 1974 for the region that now constitutes the country of South Sudan was 37.4 years. The value for this country in 2014 is not available.

(e) (i) Use the linear model to estimate the life expectancy at birth in 2014 for South Sudan.

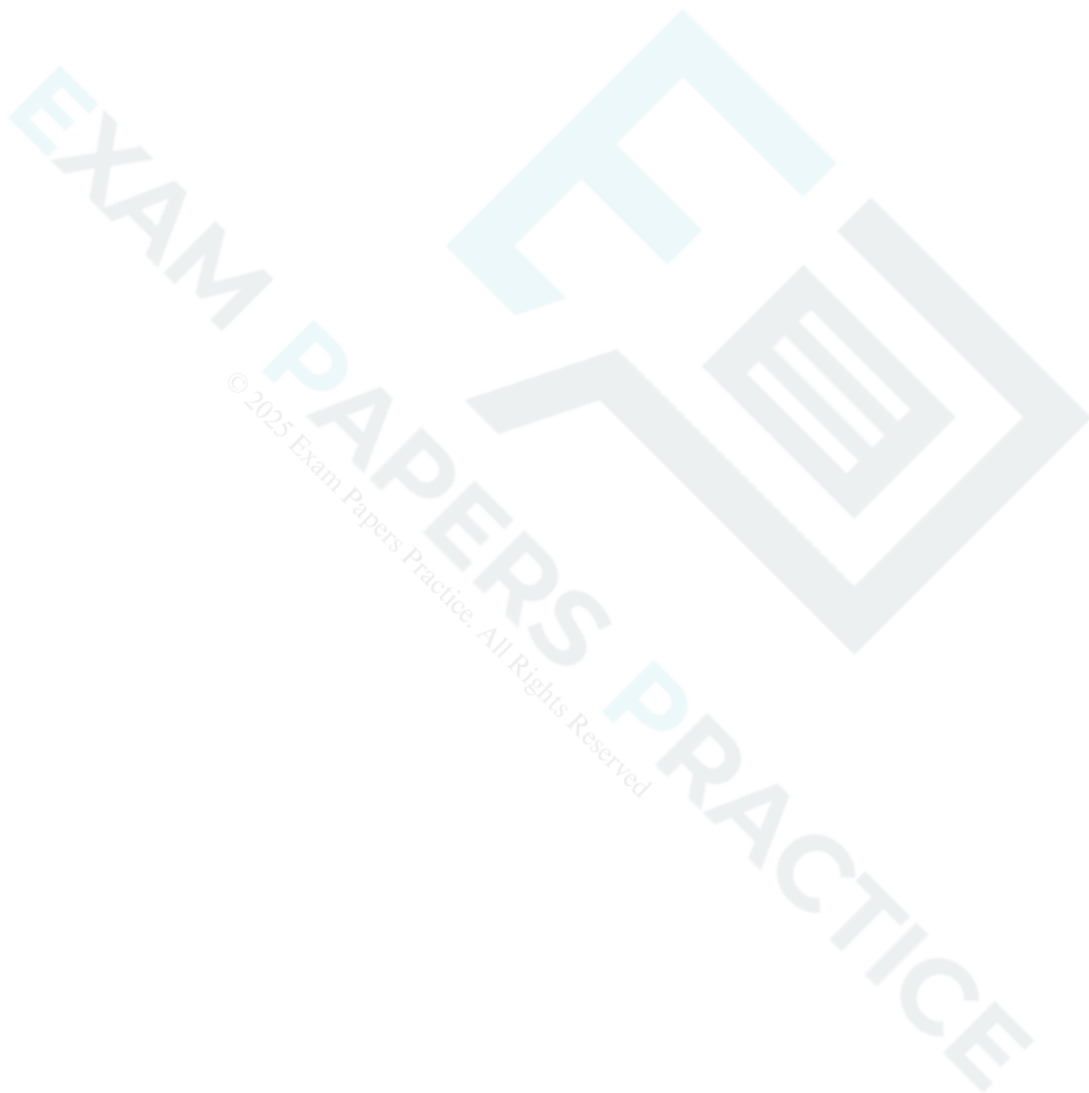
[2]

- (ii) Give two reasons why your answer to part (i) is not likely to be an accurate estimate for the life expectancy at birth in 2014 for South Sudan. You should refer to **both** information from Fig 16.4 and your knowledge of the large data set.

[2]

- (f) In how many of the countries represented in Fig. 16.4 did life expectancy drop between 1974 and 2014? Justify your answer.

[3]



- 28 (a) The graph of $y = 3\sin^2 \theta$ for $0^\circ \leq \theta \leq 360^\circ$ is shown in Fig. 6. On Fig. 6, sketch the graph of $y = 2\cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$.

[2]

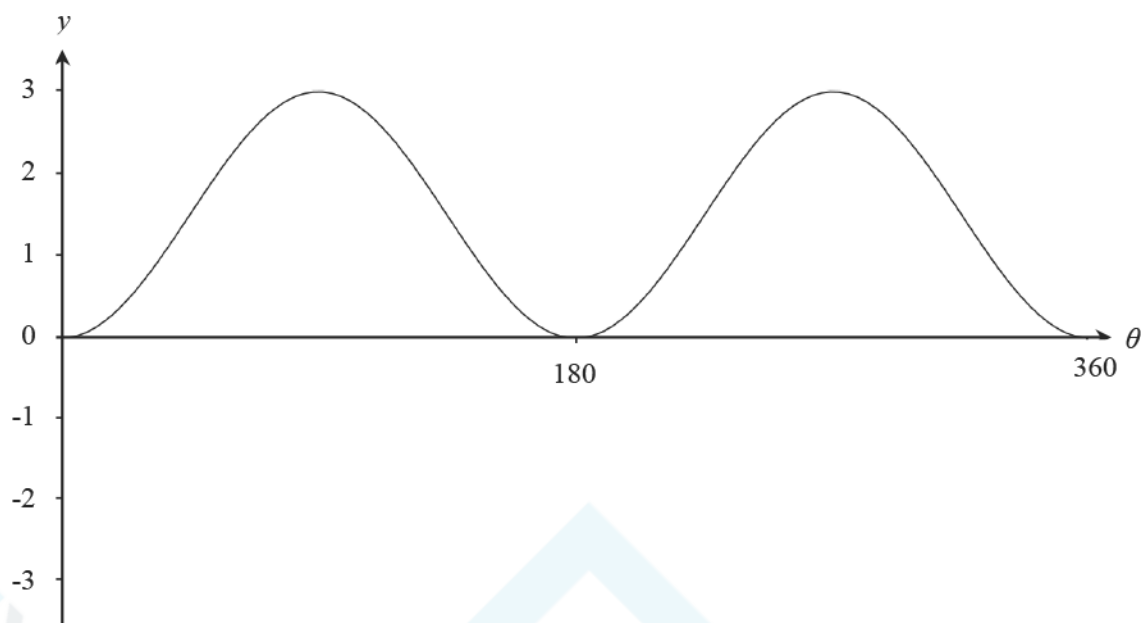


Fig. 6

(b) In this question you must show detailed reasoning.

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Determine the values of θ , $0^\circ \leq \theta \leq 360^\circ$, for which the two graphs cross.

[6]



Fig. 11 shows the curve $y = f(x)$, where $f(x)$ is a cubic function. Fig. 11 also shows the coordinates of the turning points and the points of intersection with the axes.

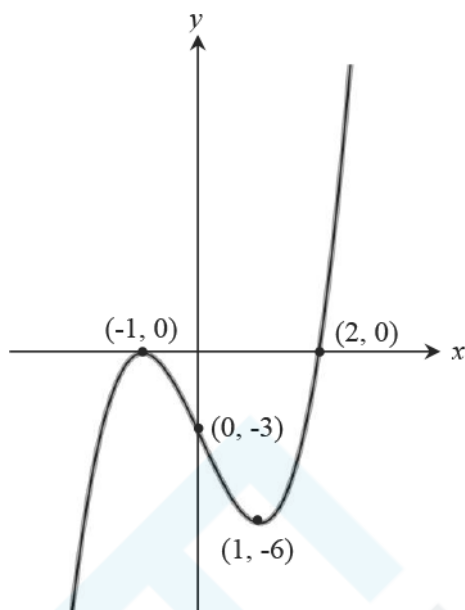


Fig. 11

Show that the tangent to $y = f(x)$ at $x = t$ is parallel to the tangent to $y = f(x)$ at $x = -t$ for all values of t .

[6]

$$x = 2\cos\theta, \quad y = \sin\theta, \quad 0 \leq \theta \leq 2\pi.$$

The point P has parameter $\frac{1}{4}\pi$. The tangent at P to the curve meets the axes at A and B.

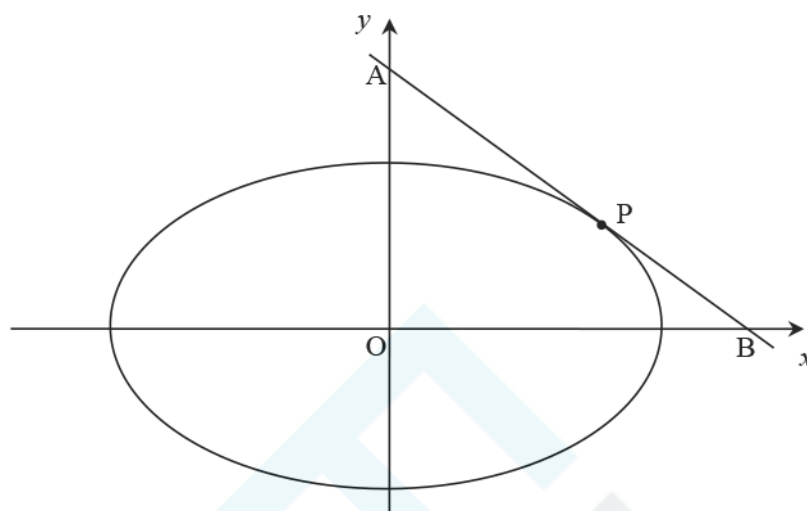


Fig.11

- (a) Show that the equation of the line AB is $x + 2y = 2\sqrt{2}$.

[6]

(b) Determine the area of the triangle AOB.

[3]

31 In this question you must show detailed reasoning.

Fig. 5 shows the circle with equation $(x - 4)^2 + (y - 1)^2 = 10$. The points $(1, 0)$ and $(7, 0)$ lie on the circle. The point C is the centre of the circle.

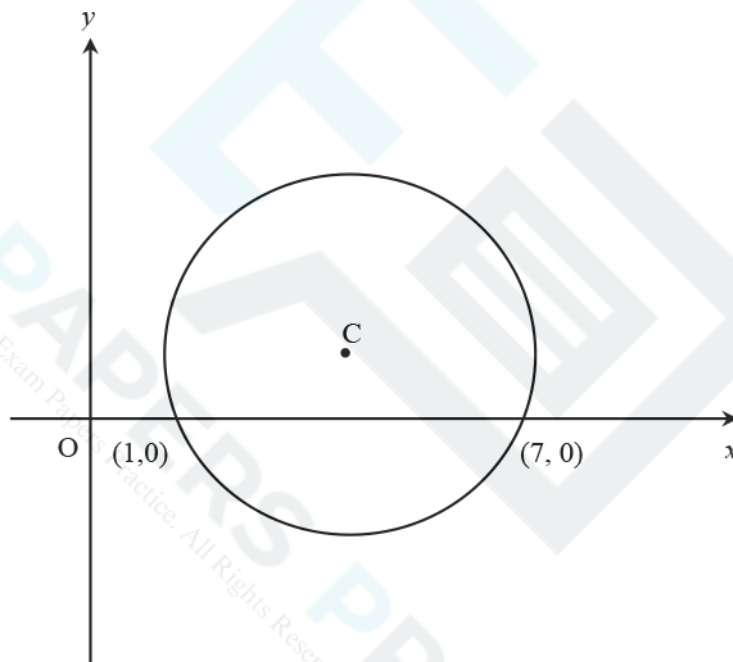


Fig. 5

Find the area of the part of the circle below the x-axis.

[5]

32 A circle has centre C $(-4, 3)$ and passes through the origin.

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(a) Find the equation of the circle.

[3]

(b) The circle crosses the positive y -axis at the point A. The tangent to the circle at A meets the x -axis at B.
Find the x -coordinate of B.

[5]

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- 33 Point P lies on the positive x -axis and point Q lies on the positive y -axis. Triangle OPQ is isosceles and its area is 18 square units. Fig. 3 shows the circle which passes through points O, P and Q.

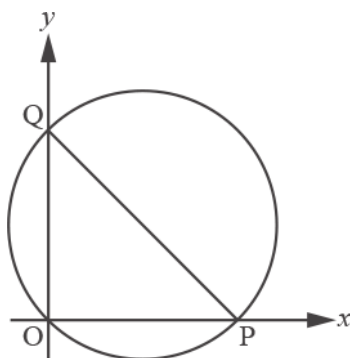


Fig. 3

Find the equation of the circle.

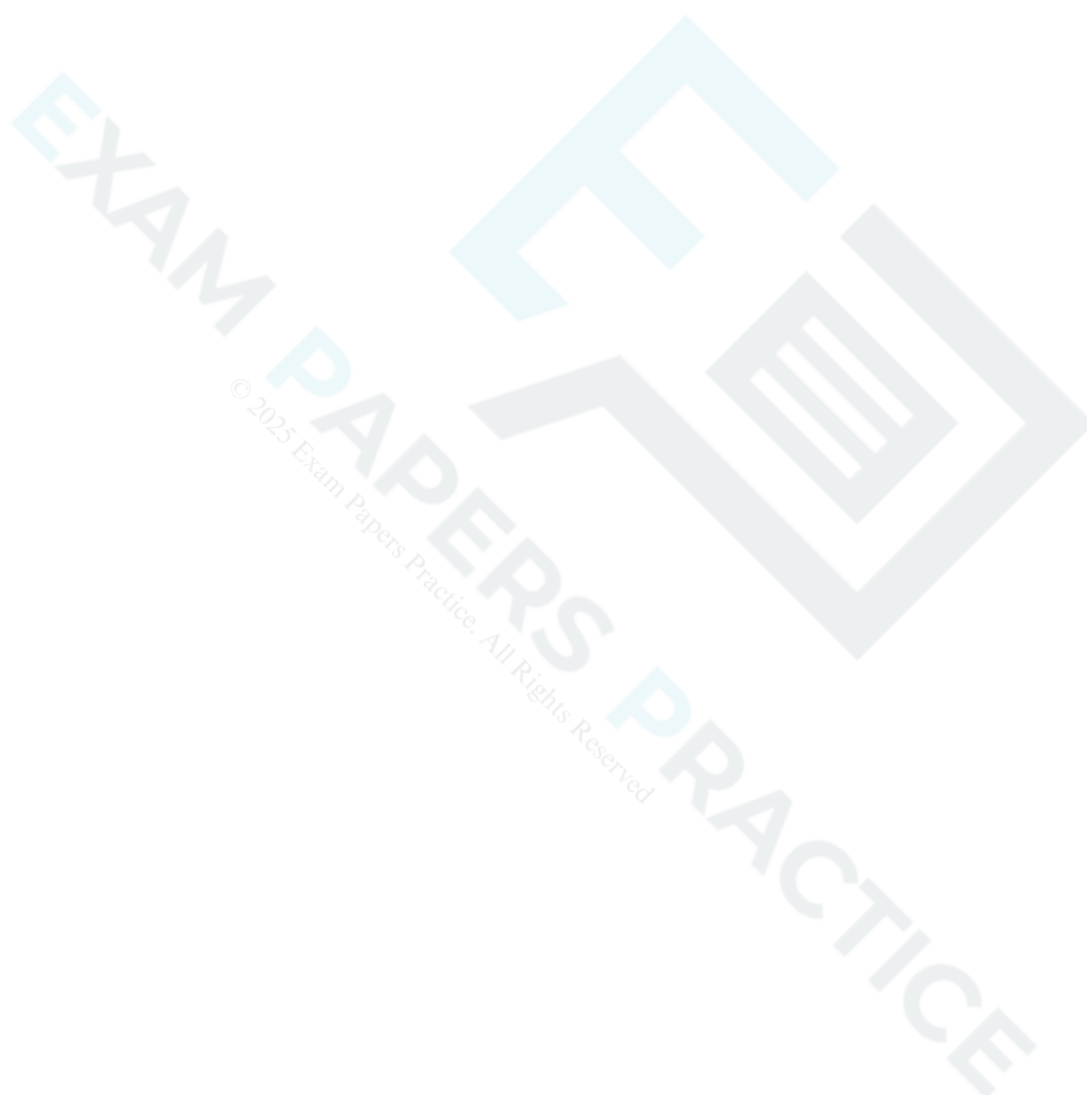
[5]

34 (a) In this question you must show detailed reasoning.

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Determine the exact values of k for which the curves $y = x^2 - kx$ and $y = 3(k + 1) + kx - x^2$ touch.

[6]



(b) Determine whether or not there is a value of k for which the curves cross on the y -axis.

[4]



35 In this question you must show detailed reasoning.

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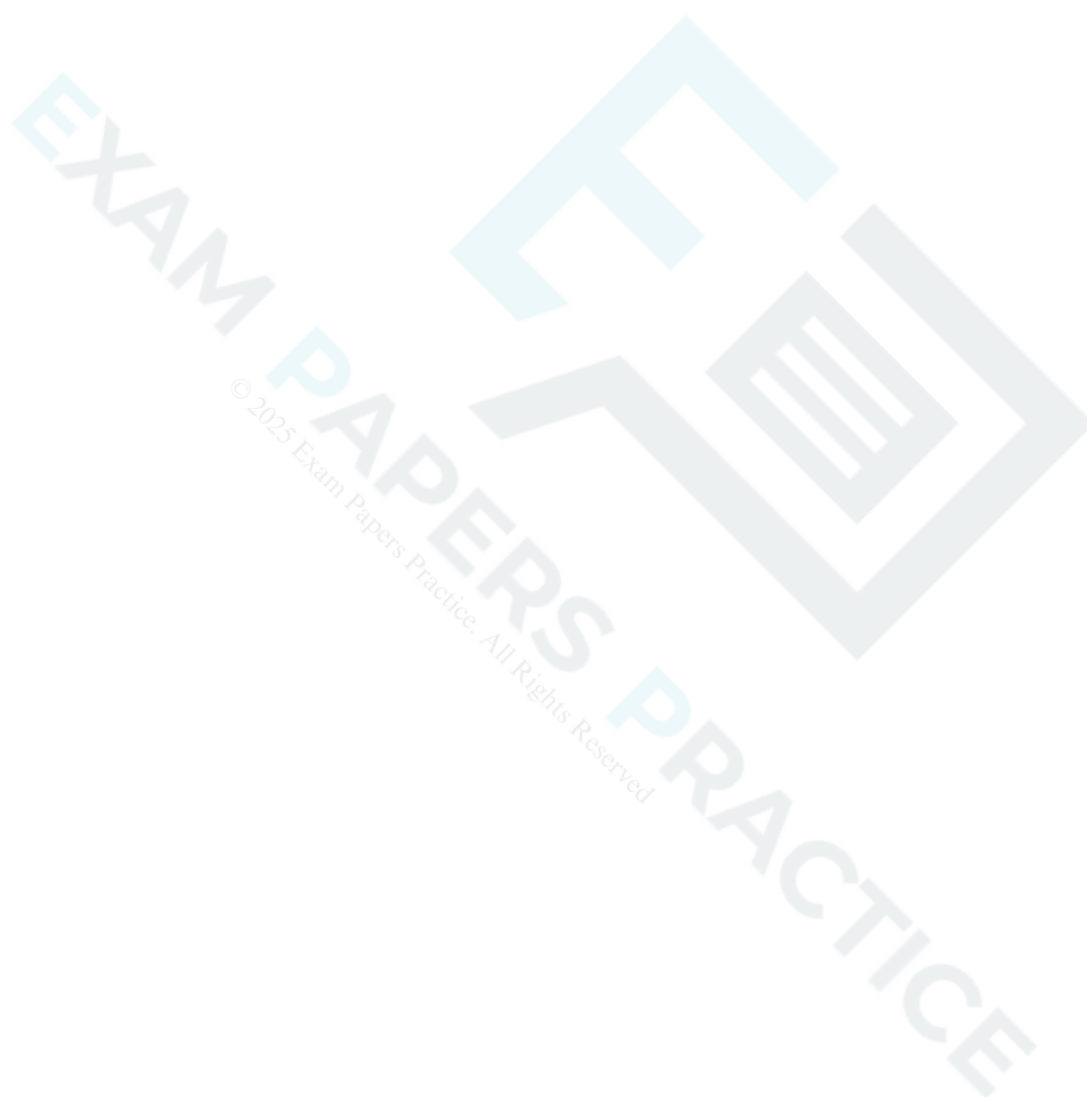
(a) For the curve $y = x^4 - 3x^2 + 2$, find the equation of the normal at the point $(1, 0)$.

[4]

(b) For the curve $y = x^4 - 3x^2 + c$, L is the normal at the point where $x = 1$. Determine the value of c for which L passes through the origin.

[3]

36 Find the equation of the straight line through (1, 5) which is perpendicular to the line with equation $2y = x + 3$. [3]



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37 In this question you must show detailed reasoning.

A curve has parametric equations

$$x = \cos t - 3t \text{ and } y = 3t - 4 \cos t - \sin 2t, \text{ for } 0 \leq t \leq \pi.$$

Show that the gradient of the curve is always negative.

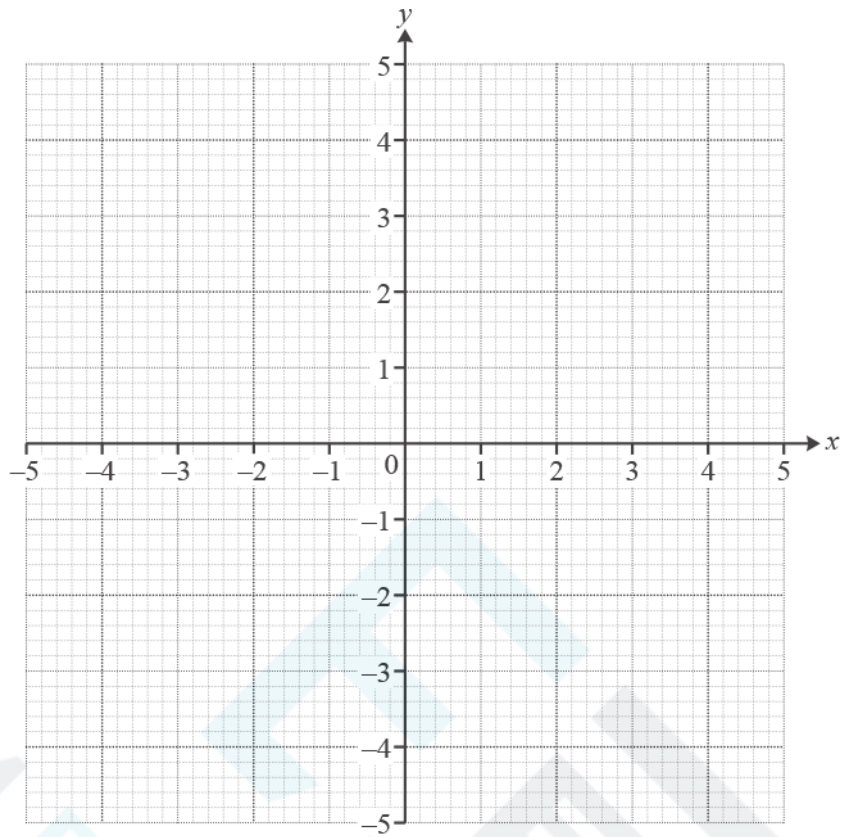
[7]



38 A straight line passes through $(0, 1)$ and has gradient -2 . Draw the graph of this line on the grid.

[2]

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39 The equation of a circle is $(x + 2)^2 + (y - 3)^2 = 5$.



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(i) State the radius of this circle and the coordinates of its centre.

[2]

(ii) Find the equation of the line through the centre of the circle which is parallel to the line $5x + y = 4$.

[2]

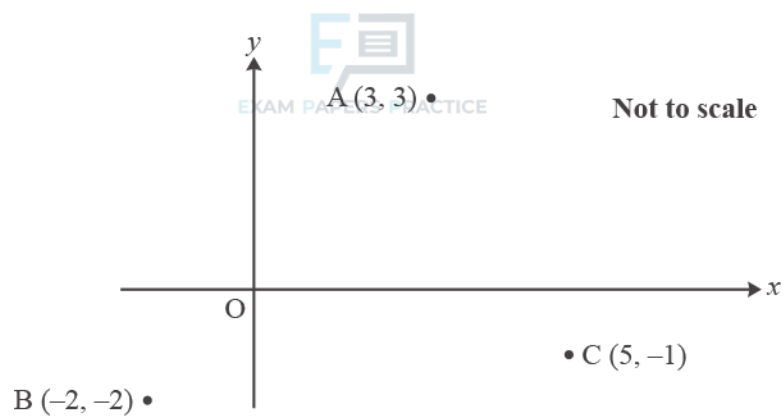


Fig. 10

Fig. 10 shows the points A (3, 3), B (−2, −2) and C (5, −1).

- (i) Show that $AB = BC$.

[2]

- (ii) Find the equation of the line through B which is perpendicular to AC. Give your answer in the form $y = mx + c$.

[4]

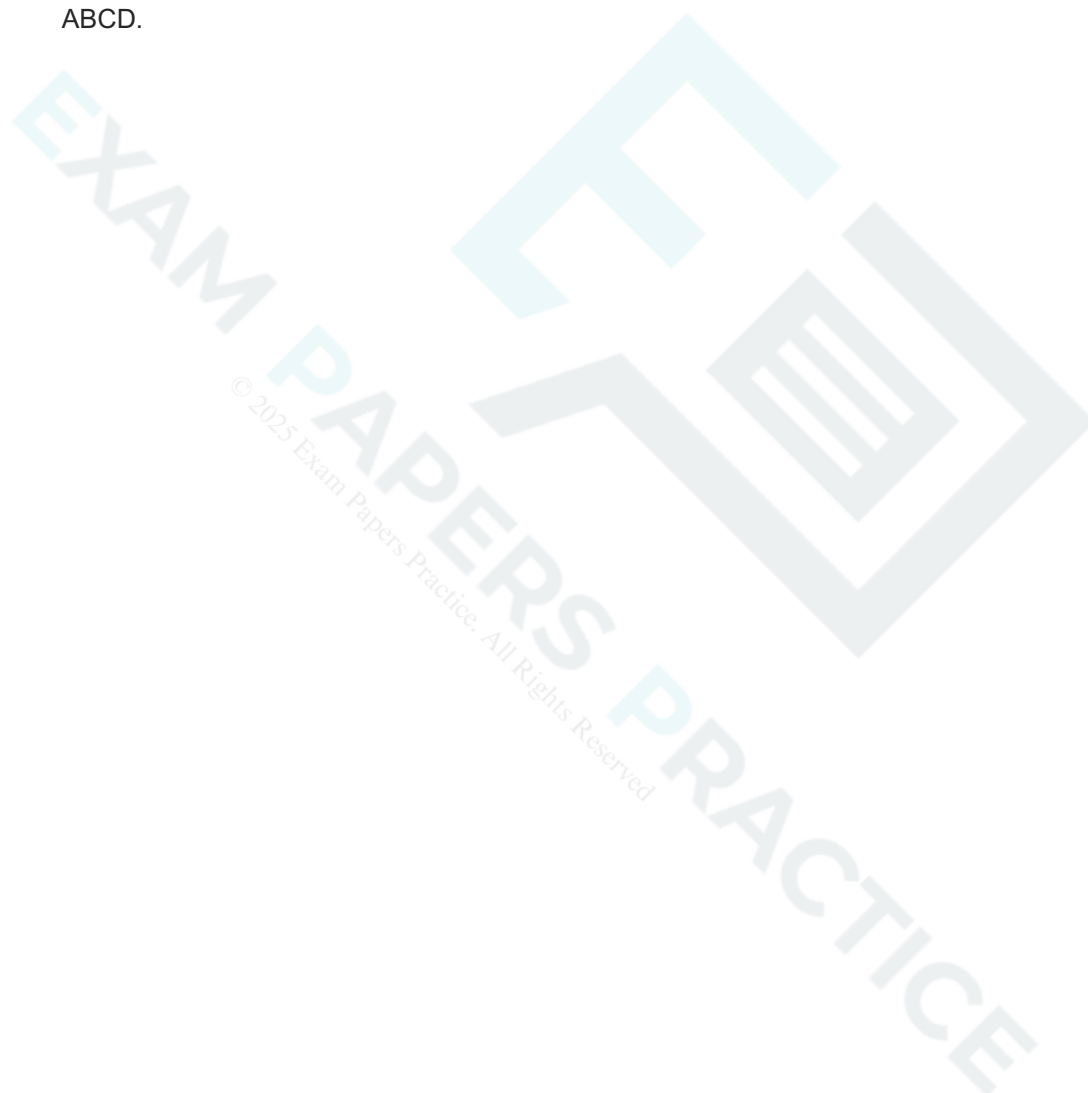
(iii) Find the coordinates of point D such that ABCD is a rhombus.

[2]

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(iv) Determine, showing all your working, whether the point E (8, 3.8) lies inside or outside the rhombus ABCD.

[4]



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41 The equation of a circle is

$$x^2 - 4x + y^2 + 6y - 12 = 0.$$

Find

(a) the coordinates of the centre of the circle,

[2]

(b) the radius of the circle.

[2]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	$\left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$ oe	B3	B1 for $a = 5/2$ oe and M1 for $6 - \text{their } a^2$ soi;	condone $\left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$ oe = 0 condone omission of index – can earn all marks bod M1 for $6 - 4.25$ or $6 - 25/2$ etc, if bearing some relation to an attempt at $6 - \text{their } 2.5^2$; M0 for just 1.75 etc without further evidence
		i	$\left(\frac{5}{2}, -\frac{1}{4}\right)$ oe or ft	B1	accept $x = 2.5, y = -0.25$ oe Examiner's Comments The majority are quite confident in the technique of completing the square, although some struggled with the arithmetic since fractions were involved. Some candidates did not complete the question and omitted to state the coordinates of the turning point; some others made sign errors such as $(-2.5, 0.25)$ after a correct completion of the square.	condone starting again and finding using calculus
		ii	(2, 0) and (3, 0)	B2	B1 each or B1 for both correct plus an extra or M1 for $(x - 2)(x - 3)$ or correct use of formula or for $\text{their } a \pm \sqrt{\text{their } b}$ ft from (i)	condone not expressed as coordinates, for both x and y values; accept eg in table or marked on graph
		ii	(0, 6)	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	graph of quadratic the correct way up and crossing both axes	B1	ignore label of their tp; condone stopping at y-axis Examiner's Comments Apart from the occasional upside down parabola and the odd cubic, most candidates made a good attempt at drawing a sketch of the curve, showing the relevant information about the intersections with the axes. They found the required factorisation straightforward, though a few candidates did resort to using the formula and in several of these cases they failed to recognise that $\sqrt{0.25}$ is equal to 0.5. The quality of the curve was often poor, probably because candidates marked the intersections on the axis first and then tried to draw the curve through them, but it was usually good enough to earn the mark.	condone 'U' shape or slight curving back in/out; condone some doubling / feathering – deleted work sometimes still shows up in scoris; must not be ruled; condone fairly straight with clear attempt at curve at minimum; be reasonably generous on attempt at symmetry
		iii	$x^2 - 5x + 6 = 2 - x$	M1	for attempt to equate or subtract eqns or attempt at rearrangement and elimination of x	accept calculus approach: $y' = 2x - 5$
		iii	$x^2 - 4x + 4 [= 0]$	M1	for rearrangement to zero ft and collection of terms; condone one error; if using completing the square, need to get as far as $(x - k)^2 = c$, with at most one error [$(x - 2)^2 = 0$ if correct]	use of $y' = -1$ M1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$x = 2, [y = 0]$	A1	condone omission of $y = 0$ since already found in (ii) if they have eliminated $x, y = 0$ is not sufft for A1 – need to get $x = 2$ A0 for $x = 2$ and another root	$x = 2$ A1
		iii	'double root at $x = 2$ so tangent' oe; www;	A1	eg 'only one point of contact, so tangent'; or showing $b^2 - 4ac = 0$, and concluding 'so tangent'; www Examiner's Comments A good number found $x = 2$ correctly. Some candidates chose to eliminate x rather than y and more often than not went wrong. Many candidates realised that a repeated root meant that the line was a tangent to the curve, but quite a few clearly did not, with some omitting the final step of showing that the line was a tangent to the curve. A small number of candidates justified the tangent by using calculus in order to determine the slope of the line and the curve at their point of intersection.	tgt is $y[-0] = -(x - 2)$ and obtaining given line A1
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2		i	midpt of AB = $\left(\frac{1}{2}, \frac{5}{2}\right)$ oe www	B2	allow unsimplified B1 for one coordinate correct	if working shown, should come from $\left(\frac{3 + -2}{2}, \frac{4 + 1}{2}\right)$ oe NB B0 for x coord. = $\frac{5}{2}$, (obtained from subtraction instead of addition)
		i	grad AB = $\frac{4 - 1}{3 - (-2)}$ oe	M1	must be obtained independently of given line; accept 3 and 5 correctly shown eg in a sketch, followed by 3/5 M1 for rise/run = 3/5 etc M0 for just 3/5 with no evidence	for those who find eqn of AB first, M0 for just $\frac{y - 4}{1 - 4} = \frac{x - 3}{-2 - 3}$ oe, but M1 for $y - 4 = \frac{1 - 4}{-2 - 3}(x - 3)$ oe ignore their going on to find the eqn of AB after finding grad AB
		i	using gradient of AB to obtain grad perp bisector	M1	for use of $m_1 m_2 = -1$ soi or ft their gradient AB M0 for just $\frac{-5}{3}$ without AB grad found	this second M1 available for starting with given line = $\frac{-5}{3}$ and obtaining grad. of AB from it
		i	$y - 2.5 = \frac{-5}{3}(x - 0.5)$ oe	M1	eg M1 for $y = \frac{-5}{3}x + c$ and subst of midpt; ft their gradient of perp bisector and midpt; M0 for just rearranging given equation	no ft for gradient of AB used

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	completion to given answer $3y + 5x = 10$, showing at least one interim step	M1	condone a slight slip if they recover quickly and general steps are correct (eg sometimes a slip in working with the c in $y = \frac{-5}{3}x + c$ - condone $3y = -5x + c$ followed by substitution and consistent working) M0 if clearly 'fudging' Examiner's Comments This part was usually done well. Most candidates were confident finding the gradient of AB, although a few failed to show their working. Almost all were then able to find the perpendicular gradient. A minority were unaware that the perpendicular bisector would pass through the midpoint of AB. Most who realised this were able to calculate the midpoint accurately. Once all the information was combined into a straight line equation, a significant minority struggled to rearrange the equation correctly because the arithmetic involved fractions. Pleasingly almost all the candidates managed to work towards the given equation, rather than trying to use the given equation to get back to a common form with their answer. Some wasted time finding the equation of AB first.	NB answer given; mark process not answer; annotate if full marks not earned eg with a tick for each mark earned scores such as B2M0M0M1M1 are possible after B2, allow full marks for complete method of showing given line has gradient perp to AB (grad AB must be found independently at some stage) and passes through midpt of AB

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$3y + 5(4y - 21) = 10$	M1	or other valid strategy for eliminating one variable attempted eg $\frac{-5}{3}x + \frac{10}{3} = \frac{x}{4} + \frac{21}{4};$ condone one error	or eg $20y = 5x + 105$ and subtraction of two eqns attempted no ft from wrong perp bisector eqn, since given allow M1 for candidates who reach $y = 115/23$ and then make a worse attempt, thinking they have gone wrong
		ii	$(-1, 5)$ or $y = 5, x = -1$ isw	A2	A1 for each value; if AO allow SC1 for both values correct but unsimplified fractions, eg $\left(\frac{-23}{23}, \frac{115}{23} \right)$ Examiner's Comments Some wasted time finding the equation of CD, which was given. Many solved the simultaneous equations correctly, but sometimes using less efficient methods, giving themselves complicated fractions to work with. A few who eliminated x struggled with simplifying $y = 115/23$. A significant minority used the implication in part (iii) that E was the midpoint of CD to obtain a solution, gaining no marks for this.	NB M0A0 in this part for finding E using info from (iii) that implies E is midpt of CD
		iii	$(x - a)^2 + (y - b)^2 = r^2$ seen or used	M1	or for $(x + 1)^2 + (y - 5)^2 = k$, or ft their E, where $k > 0$	
		iii	$1^2 + 4^2$ oe (may be unsimplified), from clear use of A or B	M1	for calculating AE or BE or their squares, or for subst coords of A or B into circle eqn to find r or r^2 , ft their E;	this M not earned for use of CE or DE or $\frac{1}{2}$ CD NB some cand's finding $AB^2 = 34$ then obtaining 17 erroneously so M0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$(x + 1)^2 + (y - 5)^2 = 17$	A1	for eqn of circle centre E, through A and B; allow A1 for $r^2 = 17$ found after $(x + 1)^2 + (y - 5)^2 = r^2$ stated and second M1 clearly earned	
		iii			if $(x + 1)^2 + (y - 5)^2 = 17$ appears without clear evidence of using A or B, allow the first M1 then M0 SC1	SC also earned if circle comes from C or D and E, but may recover and earn the second M1 later by using A or B
		iii	showing midpt of CD = $(-1, 5)$	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	showing CE or $DE = \sqrt{17}$ oe or showing one of C and D on circle	M1	alt M1 for showing $CD^2 = 68$ oe allow to be earned earlier as an invalid attempt to find r showing that both C and D are on circle and commenting that E is on CD is enough for last M1M1; similarly showing $CD^2 = 68$ and both C and D are on circle oe earns last M1M1 Examiner's Comments Most knew the form for the equation of the circle, although some used r or $\sqrt{r}$ instead of r^2. Some used C or D or the length of CD to calculate the radius, instead of using A or B. Others assumed that AB was a diameter. Very few produced enough to show that CD is a diameter, with many thinking that showing that CD is twice the radius was enough. Some stated that E was the midpoint of CD without any working to support it. This meant that the full 5 marks on this question were rarely awarded, though a significant number obtained 4 marks.	other methods exist, eg: may find eqn of circle with centre E and through C or D and then show that A and B and other of C/D are on this circle – the marks are then earned in a different order; award M1 for first fact shown and then final M1 for completing the argument; if part-marks earned, annotate with a tick for each mark earned beside where earned
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	[radius =] $\sqrt{20}$ or $2\sqrt{5}$ isw	B1	B0 for $\pm\sqrt{20}$ oe Examiner's Comments Almost all candidates obtained both marks for this part. Some gave 20 for the radius. $(-3, -2)$ was only very occasionally seen.	
		i	[centre =] (3, 2)	B1		condone lack of brackets with coordinates, here and in other questions
		ii	substitution of $x = 0$ or $y = 0$ into circle equation	M1	or use of Pythagoras with radius and a coordinate of the centre eg $20 - 2^2$ or $h^2 + 3^2 = 20$ ft their centre and / or radius	equation may be expanded first, and may include an error bod intent allow M1 for $(x - 3)^2 = 20$ and / or $(y - 2)^2 = 20$
		ii	$(x - 7)(x + 1) [=0]$	M1	no ft from wrong quadratic; for factors giving two terms correct, or formula or completing square used with at most one error	completing square attempt must reach at least $(x - a)^2 = b$ following use of Pythagoras allow M1 for attempt to add 3 to $[\pm]4$
		ii	(7, 0) and $(-1, 0)$ isw	A1	accept $x = 7$ or -1 (both required)	
		ii	$[y =] \frac{4 \pm \sqrt{(-4)^2 - 4 \times 1 \times (-7)}}{2}$ oe	M1	no ft from wrong quadratic; for formula or completing square used with at most one error	completing square attempt must reach at least $(y - a)^2 = b$ following use of Pythagoras allow M1 for attempt to add 2 to $[\pm]\sqrt{11}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$(0, 2 \pm \sqrt{11})$ or $\left(0, \frac{4 \pm \sqrt{44}}{2}\right)$ isw	A1	accept $y = \frac{4 \pm \sqrt{44}}{2}$ oe isw Examiner's Comments Finding the intersections of the circle with the axes was often well done. Almost all candidates obtained the first mark for substituting $y = 0$ or $x = 0$ in the circle equation, although some then omitted the $(-2)^2$ and/or $(-3)^2$. Some, having correctly found the x-intersections, substituted those values instead of starting again by substituting 0 to find the y values. Since the correct y equation did not factorise, there was distinctly less success in finding the y values than the x values. Some good solutions using completing the square were seen, after reaching $(y - 2)^2 = 11$, for instance, although some omitted the negative square root and then gave just one value.	annotation is required if part marks are earned in this part: putting a tick for each mark earned is sufficient
		iii	show both A and B are on circle	B1	explicit substitution in circle equation and at least one stage of interim working required oe	or clear use of Pythagoras to show AC and BC each = $\sqrt{20}$
		iii	(4, 5)	B2	B1 each or M1 for $\left(\frac{7+1}{2}, \frac{6+4}{2}\right)$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\sqrt{10}$	B2	from correct midpoint and centre used; B1 for $\pm\sqrt{10}$ M1 for $(4 - 3)^2 + (5 - 2)^2$ or $1^2 + 3^2$ or ft their centre and / or midpoint, or for the square root of this Examiner's Comments Almost all candidates were able to show that A and B lie on the circle, usually by substituting the coordinates or finding the distance between each point and the centre, though some used the longer method of substituting one coordinate and solving the resultant quadratic equation. A few candidates omitted to show that B, as well as A, lies on the circle. Almost all candidates obtained the coordinates of the midpoint of AB (4, 5) successfully, with a small minority subtracting rather than adding. Most candidates realised that the distance of the chord from the centre of the circle was the distance from (4,5) to (3,2) and obtained the correct answer of $\sqrt{10}$. Some calculated the length of AB and proceeded no further; some halved it and used Pythagoras but only a minority were successful with this approach.	may be a longer method finding length of $\frac{1}{2}$ AB and using Pythag. with radius; no ft if one coord of midpoint is same as that of centre so that distance formula / Pythag is not required eg centre correct and midpt (3, -1) annotation is required if part marks are earned in this part: putting a tick for each mark earned is sufficient
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4			substitution to eliminate one variable	M1	or multiplication to make one pair of coefficients the same; condone one error in either method	independent of first M1
			simplification to $ax = b$ or $ax - b = 0$ form, or equivalent for y	M1	or appropriate subtraction / addition; condone one error in either method	
			(0.7, 0.1) oe or $x = 0.7$, $y = 0.1$ oe isw	A2	A1 each Examiner's Comments In the main, this question was completed well. Some candidates found the arithmetic challenging, especially if rearranging $x + 3y = 1$ to substitute in for y , with the resulting need to cope with fractions. A slight majority choose the substitution method rather than elimination. A few neglected to find y having found x .	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5			$y = -0.5x + 3$ oe www isw	3	B2 for $2y = -x + 6$ oe or M1 for gradient $= -\frac{1}{2}$ oe seen or used and M1 for $y - 1 = \text{their } m$ ($x - 4$) Examiner's Comments In finding the equation of the line, most candidates obtained full marks. The main mistake was to use a gradient of 2, due to confusion between perpendicular and parallel. There was a significant number of arithmetic errors especially in coping with negative signs and the fraction $-\frac{1}{2}$.	for 3 marks must be in form $y = ax + b$ or M1 for $y = \text{their } mx + c$ and (4, 1) substituted
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	$y = (x + 5)(x + 2)(2x - 3)$ or $y = 2(x + 5)(x + 2)(x - 3/2)$	2	M1 for $y = (x + 5)(x + 2)(x - 3/2)$ or $(x + 5)(x + 2)(2x - 3)$ with no equation or $(x + 5)(x + 2)(2x - 3) = 0$ but M0 for $y = (x + 5)(x + 2)(2x - 3) - 30$ or $(x + 5)(x + 2)(2x - 3) = 30$ etc Examiner's Comments Candidates struggled with this question. Often they managed to produce the product of binomial factors $x + 2$ $x + 5$ $x - 1.5$ and failed to put it equal to y or put it equal to 0. Those who did have the correct product still very often had an expression only or equated to 0. Many candidates thought that the information about the y-intercept indicated that they should perform a vertical translation and an answer of $y = x + 2$ $x + 5$ $x - 1.5 - 30$ was fairly common among weaker students. Some candidates had an epiphany in part (ii) when they realised that their coefficients should be twice the size and sensibly went back to this part and corrected their error.	allow 'f(x) =' instead of 'y =' ignore further work towards (ii) but do not award marks for (i) in (ii)
		ii	correct expansion of a pair of their linear two-term factors ft isw	M1	ft their factors from (i); need not be simplified; may be seen in a grid	allow only first M1 for expansion if their (i) has an extra -30 etc

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	correct expansion of the correct linear and quadratic factors and completion to given answer $y = 2x^3 + 11x^2 - x - 30$	M1	must be working for this step before given answer or for direct expansion of all three factors, allow M2 for $2x^3 + 10x^2 + 4x^2 - 3x^2 + 20x - 15x - 6x - 30$ oe (M1 if one error) or M1M0 for a correct direct expansion of $(x + 5)(x + 2)(x - 3/2)$ condone lack of brackets if used as if they were there Examiner's Comments Many scored only one mark in this part, for correctly expanding a pair of their binomial factors, even after making an error in part (i). As said previously, the light dawned for many in this question and it was good to see that some of these made corrections to part (i). However, many did not and very often there would be a multiplication by 2 done at the end — with or without some attempt at justification for it.	do not award 2nd mark if only had $(x - 3/2)$ in (i) and suddenly doubles RHS at this stage condone omission of 'y =' or inclusion of '= 0' for this second mark (some cand have already lost a mark for that in (i)) allow marks if this work has been done in (i) – mark the copy of (i) that appears below the image for (ii)
		iii	ruled line drawn through $(-2, 0)$ and $(0, 10)$ and long enough to intersect curve at least twice	B1	tolerance half a small square on grid at $(-2, 0)$ and $(0, 10)$	insert BP on spare copy of graph if not used, to indicate seen – this is included as part of image, so scroll down to see it

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	-5.3 to -5.4 and 1.8 to 1.9	B2	B1 for one correct ignore the solution -2 but allow B1 for both values correct but one extra or for wrong 'coordinate' form such as (1.8, -5.3) Examiner's Comments Candidates found this straightforward on the whole, with many scoring full marks for this part. Nearly all drew an accurate line of sufficient length to intersect the curve in three places. Occasionally some read the scale incorrectly when finding the negative solution or were careless with signs, omitting the negative when writing it down.	accept in coordinate form ignoring any y coordinates given
		iv	$2x^3 + 11x^2 - x - 30 = 5x + 10$	M1	for equating curve and line; correct eqns only	annotate this question if partially correct
		iv	$2x^3 + 11x^2 - 6x - 40 [= 0]$	M1	for rearrangement to zero, condoning one error	
		iv	division by $(x + 2)$ and correctly obtaining $2x^2 + 7x - 20$	M1	or showing that $(x + 2)(2x^2 + 7x - 20) = 2x^3 + 11x^2 - 6x - 40$, with supporting working	
		iv	substitution into quadratic formula or for completing the square used as far as $\left(x + \frac{7}{4}\right)^2 = \frac{209}{16}$ oe	M1	condone one error eg a used as 1 not 2, or one error in the formula, using given $2x^2 + 7x - 20 = 0$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$[x =] \frac{-7 \pm \sqrt{209}}{4} \text{ oe isw}$	A1	dependent only on 4 th M1 Examiner's Comments There were many attempts to substitute their answers from part (iii) into the given quadratic. Many candidates did not know how to obtain this quadratic, although most eventually went on to attempt to solve it using the formula, sometimes making arithmetic errors in so doing. Of those who did attempt to derive the quadratic, there were several attempts at equating the wrong pair of equations. Some who started correctly expected to see the given answer immediately and stopped at the simplified cubic they had obtained, sometimes having an erroneous -20. Relatively few were able to show that the quadratic factor was the required one, by long division or by showing multiplying out. A very few candidates used an elegant method of equating the line and cubic and using the factorised form of each to cancel a factor of $x + 2$ on both sides before simplifying.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	$x = 4$	B1		
		i	$(4, -3)$	B1	or $x = 4, y = -3$ Examiner's Comments The minimum point was generally well found, although some just gave the y coordinate. The question said, "Write down ...", which should suggest to candidates that differentiating and putting the differential equal to zero was not needed. The line of symmetry was also usually well done, but some gave $x = -4$ or $y = 4$.	condone 4, -3
		ii	$(0, 13)$ isw	1	or [when $x = 0$], $y = 13$ isw 0 for just $(13, 0)$ or $(k, 13)$ where $k \neq 0$	annotate this question if partially correct
		ii	[when $y = 0$,] $(x - 4)^2 = 3$	M1	or $x^2 - 8x + 13 [= 0]$	may be implied by correct value(s) for x found
		ii	$[x =] 4 \pm \sqrt{3}$ or $\frac{8 \pm \sqrt{12}}{2}$ isw	A2	need not go on to give coordinate form A1 for one root correct Examiner's Comments The y intercept was usually correct. For the x intercept, many went the long way round: expanding brackets and then using the quadratic formula rather than using the completing the square method.	allow M1 for $y = x^2 - 8x + 13$ only if they go on to find values for x as if y were 0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	replacement of x in their eqn by $(x - 2)$	M1	may be simplified; eg $[y =] (x - 6)^2 - 3$ or allow M1 for $(x - 6 - \sqrt{3})(x - 6 + \sqrt{3})$ $[=0 \text{ or } y]$	condone omission of ' $y =$ ' for M1, but must be present in final line for A1
		iii	completion to given answer $y = x^2 - 12x + 33$, showing at least one correct interim step	A1	cao; condone using $f(x - 2)$ in place of y Examiner's Comments Some candidates lost a mark as they forgot that an equation has 2 sides and omitted the ' $y =$ ', only giving an expression. Most candidates realised that they should replace x with $(x - 2)$. A minority expanded brackets before replacing x with $(x - 2)$ which was a less efficient method.	
		iv	$x^2 - 12x + 33 = 8 - 2x$ or $(x - 6)^2 - 3 = 8 - 2x$	M1	for equating curve and line; correct eqns only; or for attempt to subst $(8 - y)/2$ for x in $y = x^2 - 12x + 33$	annotate this question if partially correct
		iv	$x^2 - 10x + 25 = 0$	M1	for rearrangement to zero, condoning one error such as omission of ' $= 0$ '	
		iv	$(x - 5)^2 [= 0]$	A1	or showing $b^2 = 4ac$	allow $\frac{10 \pm \sqrt{0}}{2}$ oe if $b^2 - 4ac = 0$ is not used explicitly A0 for $(x - 5)^2 = y$
		iv	$x = 5$ www [so just one point of contact]	A1	may be part of coordinates $(5, k)$	allow recovery from $(x - 5)^2 = y$
		iv	point of contact at $(5, -2)$	A1	dependent on previous A1 earned; allow for $y = -2$ found	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv				examiners: use one mark scheme or the other, to the benefit of the candidate if both methods attempted, but do not use a mixture of the schemes
		iv	for curve, $y' = 2x - 12$	M1		
		iv	$2x - 12 = -2$	M1	for equating their y' to -2	
		iv	$x = 5$, and y shown to be -2 using eqn to curve	A1		
		iv	tgt is $y + 2 = -2(x - 5)$	A1		
		iv	deriving $y = 8 - 2x$	A1		
					Examiner's Comments Since both equations were given, those were the ones which had to be used, and most candidates did so successfully. Some candidates did not realise that obtaining $(x - 5)^2 = 0$ led to sufficient evidence of a repeated root and also showed that the discriminant was zero. Some, of course, did not attempt to factorise anyway but opted for using the formula. The main error was in the very last mark, where some candidates substituted their x value back into the quadratic that they had just solved to find $y = 0$, rather than using the line or the curve to give $y = -2$.	condone no further interim step if all working in this part is correct so far
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	(7, 0)	1	accept $x = 7, y = 0$ Examiner's Comments This was usually correct, with most candidates appreciating the symmetry about $x = 4$.	condone 7, 0
		ii		2	M1 for Pythag used correctly eg $[r^2 =] 3^2 + 2^2$ or for subst A or their B in $(x - 4)^2 + (y - 2)^2 [=r^2]$ or B1 for $[r =] \pm\sqrt{13}$	annotate this question if partially correct allow recovery if some confusion between squares and roots but correct answer found
		ii	$(x - 4)^2 + (y - 2)^2 = 13$ or ft their evaluated r^2 , isw	2	M1 for one side correct, as part of an equation with x and y terms Examiner's Comments The correct method for finding the radius was usually seen; however, some candidates were let down by their poor arithmetic, for instance, $9 + 4 = 11$ or 12 or 15 or 16 . Most knew the form for the circle equation, although some failed to square the root 13 (or their number), or made some mistake in the left-hand side of the equation, such as sign errors or omitting the squared signs. Most candidates found the coordinates of D correctly using a step/vector method. A few tried to find the intersection of AD with the circle, but these were usually unsuccessful.	do not accept $(\sqrt{13})^2$ instead of 13; allow M1 for LHS for $(x - 4)^2 + (y - 2)^2 = r^2$ (or worse, $(x - 4)^2 + (y - 2)^2 = r$) (may be seen in attempt to find radius)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	(7, 4)	2	B1 each coord accept $x = 7$, $y = 4$ if B0, then M1 for a vector or coordinates approach such as '3 along and 2 up' to get from A to C oe or M1 for $\frac{x_D + 1}{2} = 4$ and $\frac{y_D + 0}{2} = 2$ Examiner's Comments Attempts at this part were variable. Whilst many recognised that they had to find the gradient of AD first, quite a few made errors in doing this. Most recognised that the gradient of the tangent was the negative reciprocal of this number and substituted this into either $y - y_1 = m(x - x_1)$ or $y = mx + c$. A few used their gradient of AD in the equation of the line. Quite a number of errors were seen in reaching the final answer, most of these associated with dealing with fractions. A few candidates did not give their answer in the required form.	condone 7, 4 or M1 for longer method, finding the equation of the line CD as $y = \frac{2}{3}(x - 1)$ oe and then attempting to find intn with their circle
		iv	grad tgt = $-\frac{3}{2}$ oe	M2	correctly obtained or ft their D if used M1 for grad AD = $\frac{4 - 0}{7 - 1}$ oe isw or $\frac{2}{3}$ oe seen or used in this part or for their grad tgt = $-1/$ their grad AD	annotate this question if partially correct may use AD, CD or AC NB grad AD etc may have been found in (iii); allow marks if used in this part – mark the copy of (iii) that appears below the image for (iv)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$y - \text{their } 4 = \text{their } (-3/2)(x - \text{their } 7)$	M1	or subst (7, 4) into $y = \text{their } (-3/2)x + b$	
		iv	$y = -1.5x + 14.5$ oe isw	A1	M0 if grad AD oe used or if a wrong gradient appears with no previous working must be in form $y = ax + b$	condone $y = \frac{-3x + 29}{2}$ condone $y = -1.5x + b$ and $b = 14.5$ oe
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9			midpt M of $AB = \left(\frac{1+6}{2}, \frac{5-1}{2} \right)$ oe isw soi	1	condone lack of brackets; accept in the form $x = 7/2$ oe, $y = 2$ oe	
			subst of their midpt into $y = 2x - 5$ and attempting to evaluate	1	eg $2 \times \text{their } 3.5 - 5 = \text{their}$ result accept $2 = 2 \times 3.5 - 5$	alt methods: allow 2 nd M1 for finding correct eqn of AB as $y = -\frac{6x}{5} + \frac{31}{5}$ oe and attempting to solve as simult eqn with $y = 2x - 5$ for x or y or allow M1 for finding in unsimplified form the eqn of the line through their midpt with gradient 2 and A1 for showing it is $y = 2x - 5$, so Yes
			all work correct and 'Yes' oe	1	Examiner's Comments Many obtained three marks here without any difficulty, with many candidates choosing to use the quick substitution of midpoint method to prove that the point was on the line. A minority failed to state a clear conclusion once this step had been performed. Longer methods were seen occasionally but were rarely completed successfully, with the equation of AB sometimes being found simply because the candidate did not know what to do.	
Total				3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	[Grad AB =] $\frac{7-3}{2-0}$ or $\frac{4}{2}$	M1		
		i	[Grad BC =] $\frac{-1-3}{8-0}$ or $\frac{-4}{8}$	M1		allow just a simplified version of 2 or $-1/2$ for one method mark, but for both to be gained, there must be evidence that the gradients have been obtained independently
		i	product of gradients = -1 [when lines are at right angles]	A1	or 'negative reciprocal [so perpendicular] oe; may be implied by correct calculation	may be seen earlier, but correct working must support the statement
		i	or $AB^2 = 2^2 + 4^2 [=20]$, $BC^2 = 8^2 + 4^2 [=80]$ and $AC^2 = 6^2 + 8^2 [= 100]$	M2	or equiv for AB etc; allow at unsimplified stage; or M1 for just one correct expression for one of the sides	allow just a simplified version of eg $AB^2 = 20$ for one method mark, but for both to be gained, there must be evidence that the lengths or their squares have been obtained independently
		i	$AB^2 + BC^2 = AC^2$ [so by Pythagoras, angle ABC = 90°] oe	A1	may be implied by correct calculation	may be seen earlier, but correct working must support the statement

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i				another possible method: M1 for finding midpt of AC as (5, 3), M1 for showing dist from midpt to A, B and C is 5 and M1 for using angle in a semicircle to show that $\angle ABC = 90^\circ$ Examiner's Comments The gradient method was the most common. Most candidates showed how their gradients were obtained – it was not sufficient to quote 2 and $-\frac{1}{2}$, since for proof, evidence was needed that the gradients were independently obtained. Some candidates did not obtain the final mark, merely saying that the gradients are perpendicular; a reason was needed, for instance stating that $2x - \frac{1}{2} = -1$, or stating that the gradients are negative reciprocals of each other. The other popular method was to calculate the lengths of the sides of triangle ABC and use Pythagoras's theorem, which was usually well done, although some candidates sadly confused squares and square roots, such as stating $\sqrt{20} + \sqrt{80} = \sqrt{100}$.
		ii	centre $D = \left(\frac{2+8}{2}, \frac{7+(-1)}{2} \right)$ or (5,3) so i	B1	may be implied by circle eqn	if already found in (i), must be used in (ii) to get the mark here
		ii	radius = 5 or $r^2 = 25$ or for finding dist between A, B or C and their centre D oe	B1	may be implied by circle eqn	if already found in (i), must be used in (ii) to get the mark here

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$(x - a)^2 + (y - b)^2 = r^2$ soi	M1	general formula may be quoted or implied by eqn using their values, but it must be clear that they are using their r^2 rather than their r or their d or d^2	for this method mark, allow use of their values, even if obtained from AB or BC as diameter
		ii	$(x - 5)^2 + (y - 3)^2 = 25$ or 5^2 isw	A1	alternative method: allow B4 for $(y = 7)(y + 1) + (x - 2)(x - 8) = 0$	Examiner's Comments The centre was usually found correctly. The radius caused more problems with some calling AC the radius instead of the diameter, and others reaching $\sqrt{100} / 2$ but then making an error. The form of the circle equation was not always correct, with sign errors seen on the left-hand side as well as the right-hand side sometimes being r, d, or d^2 instead of the correct r^2.
		iii	[grad AD =] $\frac{7-3}{2-5}$ isw or $-\frac{4}{3}$ oe	B1	$CD \left(\frac{-1-3}{8-5} \right), AC \left(\frac{7-(-1)}{2-8} \right)$ or ft their D from (ii) or B1 for correct differentiation: $2x + 2y \frac{dy}{dx} - 10 - 6 \frac{dy}{dx} = 0$ oe	if D wrong, check back to (ii) for any ft NB: A(2, 7) B(0, 3) and C(8, -1)
		iii	grad tgt = $\frac{3}{4}$ oe www or -1 / their grad AD oe	M1		
		iii	$y - 7 =$ their $\frac{3}{4}(x - 2)$ or $7 =$ their $\frac{3}{4} \times 2 + c$	M1	M0 if grad AD used; M0 for a spurious gradient used	perp gradient to AB or BC used: may earn 2nd M1 only

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$4y = 3x + 22$ oe where a, b, c are integers, isw	A1	allow correct answer to imply 3rd M1, provided first two Ms have been earned	Examiner's Comments Finding the gradient of the tangent was generally well done, although some used the gradient of AB or the perpendicular to AB as the gradient of the tangent. Some lost the final mark by not writing the equation in the requested form with integer coefficients. A few weak candidates had no idea how to proceed with this part and omitted it.
			Total	11		
11			$\text{gradient} = \frac{4\sqrt{9.5} - 12}{9.5 - 9}$ 0.6577 to 0.66 $9 < x_c < 9.5$	M1 A1 B1	or 0.657656...isw Examiner's Comments This was done very well. Some candidates lost the second mark through premature rounding or simply giving the answer as 0.6. Only a few calculated the reciprocal of the gradient (which didn't score) and nearly all gave an appropriate value for x_c . A few candidates differentiated and substituted values in the derivative.	$4\sqrt{38} - 24$ $4\sqrt{38} - 24$ allow $8.53 \leq x_c < 9$
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	ruled line touching curve at $x = 2$	M1		intent to touch, but must not clearly cut curve
		i	their $\frac{y_2 - y_1}{x_2 - x_1}$ from their <i>tangent</i>	M1	may be on graph or in working; must use correct points from their line their tangent may be at another point	M0 for reciprocal
		i	answer in range 2.5 to 3.0 inclusive	A1	both M1s must be awarded <u>Examiner's Comments</u> More than half the candidates earned full marks; only a small minority failed to score at all. Almost all candidates drew a reasonable tangent, though it was occasionally at (1, 2) instead of (2, 4). A few lines were not tangents at all, the normal being the usual error, although occasionally curves were seen. Most candidates knew that they should draw a right-angled triangle, but many were very small, leading to a gradient which was outside the acceptable range. Some candidates clearly used two points taken from the curve which did not score, others tried (vainly) to differentiate the function, perhaps not understanding the word 'hence' in the question.	(value is approx 2.773)
		ii	3.482202253... and 4.59479342... rot to 3 or more sf	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	2.78 to 2.7815 or 2.8	B1	mark the final answer <u>Examiner's Comments</u> Approximately one third of candidates scored full marks and nearly all knew what was required. However, marks were commonly lost because of premature approximation. Candidates whose values for 2^x were 3.5 and 4.6 calculated a gradient of 2.75, outside the range and earned no marks. Candidates who stated the values 3.48 and 4.59 earned the first mark, but lost the second if they left their gradient as 2.775 rather than correcting it to 2.8 or 2.78, all that their values were qualified to give. Candidates who gave more figures (up to ten) usually earned the second mark. A few candidates calculated $\frac{\Delta x}{\Delta y}$; a few calculated the midpoint or calculated the gradient using the point (2, 4).	2.781477917..
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$\left[\frac{dy}{dx}\right] 4 \times 2 + 3$ or 11 isw	M1*		
		i	$9 = \text{their } (4 \times 2 + 3) \times 2 + c$	M1dep*	or $y - 9 = \text{their } (4 \times 2 + 3) \times (x - 2)$	
		i	$y = 11x - 13$ or $y = 11x + c$ and $c = -13$ stated isw	A1	or $y - 9 = 11(x - 2)$ isw Examiner's Comments The majority of candidates gained full marks on this question. A few found the gradient correctly and then went on to find the equation of the normal, and some candidates integrated and found the equation of the curve.	
		ii	$\frac{4x^2}{2} + 3x$	M1*		
		ii	$[y =] 2x^2 + 3x + c$	A1	must see "2" and "+ c", may be earned later eg after attempt to find c	
		ii	$9 = 2 \times 22 + 3 \times 2 + c$	M1dep*	must include constant, which may be implied by answer	
		ii	$y = 2x^2 + 3x - 5$ cao	A1	allow first 4 marks for $y = 2x^2 + 3x + c$ and $c = -5$ stated	
		ii	$(1, 0)$ and $(-2.5, 0)$ oe cao	B1	or for $x = 1, y = 0$ and $x = -2.5, y = 0$	B0 for just stating $x = 1$ and $x = -2.5$
		ii	$x = -\frac{3}{4}$	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y = -\frac{49}{8}$	B1	–6.125 or – 6$\frac{1}{8}$ Examiner's Comments Most candidates integrated successfully. A few omitted '+ c' and made little progress thereafter, but the majority successfully obtained the equation of the curve. Many candidates failed to give the coordinates in a correct form or transposed the signs, thus losing an accuracy mark. Most candidates used the given derivative to find the co-ordinates of the minimum point, but it was surprising how many made a sign error and then couldn't obtain the correct value for y. A significant number of candidates completed the square instead of using the derivative, and most lost accuracy.	
		iii	substitution to obtain [y =] f(2x) in polynomial form	M1	f (x) must be the quadratic in x with linear and constant term obtained in part (ii), may be in factorised form	or their x = 1 → their 0.5 and their x = – 2.5 → their x = –1.25
		iii	$y = (2x - 1)(4x + 5)$ or $y =$ or $y = 2\left(2x + \frac{3}{4}\right)^2 - \frac{49}{8}$	A1FT	must be simplified to one of these forms, FT their quadratic in x with linear and constant term obtained in part (ii)	hence $y = (2x - 1)(4x + 5)$ FT their x-intercepts from their quadratic in x with linear and constant term obtained in part (ii)

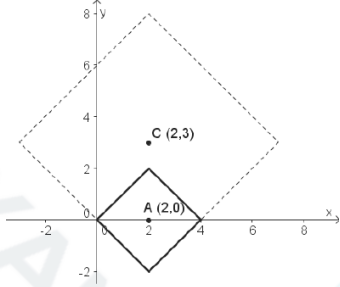
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\left(-\frac{3}{8}, -\frac{49}{8}\right)$ oe	B1	or FT their (both non-zero) co-ordinates for minimum point or their quadratic in x with linear and constant term obtained in part (ii) Examiner's Comments Very few candidates realised that they needed to work with $f(2x)$ to find the new equation. The majority of those who did adopt the correct approach often went wrong, usually with the first term. In order to earn the method mark by this approach examiners needed to see the substitution: many candidates just wrote down $y = 4x^2 + 6x - 5$ and failed to score. It may have been the case that the correct approach was being attempted, but this answer was also seen resulting from totally wrong working! A few candidates successfully worked with the images of the intercepts following the stretch, but often failed to simplify the answer correctly. A minority of candidates earned the third mark, either as a follow through mark or for a fully correct answer. However, many candidates multiplied the x value by 2, or halved the y value instead or as well.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$\frac{(5.1^2 - 10.2) - (5^2 - 10)}{5.1 - 5}$ oe	M1	condone omission of brackets	0 for 8.1 unsupported
		i	8.1	A1	Examiner's Comments The majority of candidates gained full marks on this question. A significant minority differentiated and substituted in the midpoint, or the endpoints of the chord and found the mean. Whilst these approaches do achieve the correct numerical answer, they nevertheless went unrewarded.	
		ii	$\frac{(5+h)^2 - 2(5+h) - \text{their } 15}{h}$ oe	M1	condone omission of brackets	
		ii	$25 + 10h + h^2 - 10 - 2h$ oe seen	M1	allow one sign error	
		ii	numerator is $8h + h^2$	A1		
		ii	$8 + h$ isw	A1	Examiner's Comments Many candidates clearly didn't understand the notation, and either produced expressions involving x and h , or "expanded brackets" and worked with $5f + fh$. A good number of candidates did understand what this question was about, and successfully substituted to obtain correct expressions. Some made sign errors or slips in arithmetic: $h + 12$ was a common wrong answer, and a few knew what the answer was supposed to be and "back-engineered" their incorrect work accordingly.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$h \rightarrow 0$	M1	may be embedded; allow eg "tends to 0"	M0 for differentiation of $x^2 - 2x$ M0 for following from part (i) M0 for $h = 0$
		iii	their 8	A1	FT their $k + h$ from part (ii) Examiner's Comments Only a few candidates used the correct terminology or notation here. Some worked with $h = 0$ and a good number ignored part (ii) and differentiated. Neither approach scored.	
		iv	$y = 8x - 25$ isw	B1	or $y - 15 = 8(x - 5)$ isw or $y = 8x + c$ and $c = -25$ stated isw	
		iv	non-zero numerical value for x-intercept on their straight line found	M1		
		iv	$[x =] 3.125$ oe	A1	may be embedded in calculation for area	
		iv	$\frac{1}{2} \times$ their non-zero y-intercept $\times$ their $\frac{25}{8}$	M1	condone arithmetic slips in finding values of intercepts	or integration and evaluation of their $\int_0^{25/8} (8x - 25)dx$; lower limit must be 0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$\frac{625}{16}$ or $39\frac{1}{16}$ or 39.0625 isw	A1	accept rounded to 1 dp or better for A1; but A0 if final answer negative Examiner's Comments Many candidates found the correct equation and went on to achieve full marks. Some didn't read the question carefully and used (5, 15) with (3.125, 0). A small number of candidates found the equation of the normal and were thus only able to access two method marks.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	B must lie on one of the two axes	B1	oe, say, horizontal and vertical lines from (0,0).	Do not accept a list of points unless an overall statement that includes all points is given (not just integers).
		ii	B lies on line $y = x$	B1	oe	As (i)
		ii	or on line $y = -x$	B1	Examiner's Comments There were many correct solutions using mathematical terms such as the x-axis, y-axis, $x=0$, $y=0$, $y=x$ and $y=-x$. Some others gave appropriate descriptions such as all points $(p,0)$ for all real values p or said all points vertically above and below and horizontally from A or equivalent whilst others used the points of the compass in their descriptions. Some only gave a list of points which was insufficient without a general statement to include all points. Some only gave integer values. In several cases one felt that the candidates probably did know the correct answer but were unable to explain it clearly.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$(-0.7, 5.3)$	B1	Examiner's Comments There were some wrong answers with unclear methods although there were also many correct solutions.	SC B1 for $(y =)$ 5.3 and 0.7.
		i	$(-0.7, 0.7)$	B1		
		ii		B1	A square with straight edges as shown with a vertex at (2,2). (and nothing more)	
		ii			Examiner's Comments This part was usually right.	
			Total	3		
17			The 10 routes shown from (0,0) to (3,2) each continue in one way via (4,2) to (4,3) and each continue in one way via (3,3) to (4,3) Hence 20	2 B2	www Need explanation SC B1 without explanation Examiner's Comments This was less successful. Many gave the answer 10 as $n(3,2)$ is 10. Others gave $10+2=12$.	Or, the 35 all pass through either 20(3,3) or 15 (4,2). 10 of the 20 at (3,3) come from (3,2) [and the rest from (2,3)] 10 of the 15 at (4,2) come from (3,2) [and the rest from (4,1)] So $10 + 10 = 20$ are from (3,2) oe
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18			Evidence of $35 + n(3,4)$ 70	M1 A1	70 www gets B2. Examiner's Comments Many obtained both marks although some left their answer as $n(3,4)+n(4,3)$ without evaluation.	
			Total	2		
19		i	$t(P,A) = t(P,B)$	B1	(not $t(P,A) = t(P,B) = 3$) Examiner's Comments This was usually correct although some stated $t(P,A)=t(P,B)=3$ which was wrong.	Accept words, such as, [distance] $t(P,A)$ is equal to [distance] $t(P,B)$.
		ii	$t(P,A) = 3$	B1	Examiner's Comments $t(P,A)=3$ was usually given.	Accept words such as, all the points with [distance] $t(P,A)$ equal to 3.
			Total	2		



Question			Answer/Indicative content	Marks	Part marks and guidance	
20				B3	B3: Exactly as shown (with line extended beyond (4,2) and before (0,3)) Award B1 for a locus including one partially correct line segment or at least two correct discrete points in that segment Award B2 for a locus that contains parts of all three correct line segments (or at least two discrete points within each of them) and no additional incorrect points. Examiner's Comments This was generally well answered although some just gave one point. Some got the sloping part right but had vertical, not horizontal, lines. Others had the right horizontals but the sloping line was flatter.	
			Total	3		
21			(4,0) and (5,1) and (2,4)	B3	All three points correct (and no additional points) Accept a list of coordinates. Examiner's Comments Most candidates scored all three marks.	SC B1 any correct point SC B2 any two correct points (and no more than two incorrect additional points) B0 if points unclear
			Total	3		

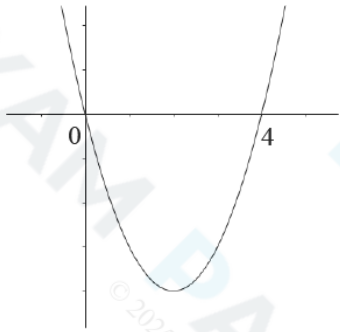
Question			Answer/Indicative content	Marks	Part marks and guidance		
22			Midpoint of $AB = (\frac{5}{2}, 0)$ Gradient of perpendicular to $AB = -\frac{1}{2}$ Perpendicular bisector equation is $y = -\frac{1}{2}(x - \frac{5}{2})$ oe Position vector is $\frac{1}{2}\mathbf{i} + \mathbf{j}$ Alternative method Suppose C has position vector $\mathbf{c} = p\mathbf{i} + \mathbf{j}$ $\overrightarrow{AC} = (p-1)\mathbf{i} + 4\mathbf{j}$ oe or $AC ^2 = (p-1)^2 + 4^2$ oe $\overrightarrow{BC} = (p-4)\mathbf{i} - 2\mathbf{j}$ oe or $BC ^2 = (p-4)^2 + 2^2$ oe $(p-1)^2 + 4^2 = (p-4)^2 + 2^2$ Position vector is $\frac{1}{2}\mathbf{i} + \mathbf{j}$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) A1(AO1.1) [4] M1(AO3.1a) M1(AO1.1) A1(AO1.1) A1(AO1.1)			
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
23		a	Radius = 5 (2, -3)	B1(AO1.1) B1(AO1.1) [2]			
		b	$(y + 3)^2 = 21$ or $y^2 + 6y - 12 = 0$ Roots $-3 + \sqrt{21}$ and $-3 - \sqrt{21}$ $(0, -3 + \sqrt{21})$ and $(0, -3 - \sqrt{21})$	M1(AO1.1a) A1(AO1.1) A1(AO1.1) [3]	Substituting $x = 0$ and rearranging For one y -value All correct		
		c	$(1 - 2)^2 + (2 + 3)^2 = 1^2 + 5^2$ E.g. This is more than 25 so outside the circle	M1(AO1.1) A1(AO1.1) [2]	Or distance of (1, 2) from their centre Or distance of (1, 2) from centre is $\sqrt{26} > 5$, so outside the circle		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	Gradient $CP = \frac{1 - (-3)}{-1 - 2} = -\frac{4}{3}$ FT their C(entre) Gradient of tangent $= \frac{3}{4}$ FT their grad CP Equation of tangent $y - 1 = \frac{3}{4}(x - (-1))$ FT their grad $4y = 3x + 7$ oe	M1(AO1. 1a) M1(AO1. 1) M1(AO1. 1) A1(AO1. 1) [4]			
			Total	11			

Question			Answer/Indicative content	Marks	Part marks and guidance		
24		a	Correct shape for $y = \frac{1}{x}$, translated vertically upward Crosses x-axis at $x = -\frac{1}{a}$ Asymptotes $x = 0$ and $y = a$	B1(AO1.1) B1(AO2.2a) B2(AO1.1) (AO2.2a) [4]	B1 for one asymptote correct		
		b	$\frac{dy}{dx} = -\frac{1}{x^2}$ When $x = 2$, gradient = $-\frac{1}{4}$ Gradient of normal = 4 FT their gradient $y - \frac{5}{2} = 4(x - 2)$ FT their gradient $2y = 8x - 11$ oe	M1(AO1.1a) M1(AO1.1) M1(AO2.1) M1(AO1.1) A1(AO2.1) [5]	Or gradient of given line is 4 or $y = \frac{5}{2}$ at the point on the curve where $x = 2$ At least one correct interim step or clear check that $\left(2, \frac{5}{2}\right)$ is on given line Any simplified form		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$2\left(\frac{1}{x} + 2\right) = 8x - 11$ $8x^2 - 15x - 2 = 0$ Other point is $x = -\frac{1}{8}$, $y = -6$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Substitution Forming quadratic, condone one error BC $x = 2$ not needed in this case		
			Total	12			

Question		Answer/Indicative content	Marks	Part marks and guidance		
25		DR $2x^2 - kx = x^2 - k$ $x^2 - kx + k = 0$ discriminant = $k^2 - 4k$ $k^2 - 4k \geq 0$  $k \leq 0$ or $k \geq 4$	B1(AO3.1a) M1(AO2.1) B1(AO1.2) M1(AO1.1) M1(AO2.4) A1(AO2.5) [6]	Equating the two expressions must be seen Condone one error in rearranging Or give table of values, or or $\{k : k \leq 0\} \cup \{k : k \geq 4\}$		
Total			6			

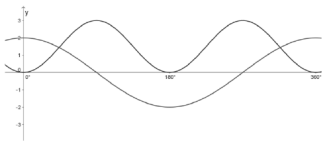
Question		Answer/Indicative content	Marks	Part marks and guidance		
26		DR $y = 4 - 2x$				
		$4 - 2x = x^2 + x$	M1(AO2.1)	Eliminating x or y must be seen		
		$\Rightarrow x^2 + 3x - 4 = 0$	M1(AO1.1)	Form a quadratic equation	Or $y^2 - 14y + 24 = 0$	
		$\Rightarrow x = 1$ or $x = -4$	A1(AO1.1)		SC1 for one pair of coordinates only	
		$y = 2$ or $y = 12$	A1(AO1.1)			
		$(1,2)$ and $(-4,12)$	A1(AO2.5)			
			[5]	For final A mark, corresponding values of x and y must be expressed as coordinates from well set out correct solution		
		Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
27		a	Comment about shape of distribution for first graph Comment about shape of distribution for second graph	B1(AO2.2b) B1(AO2.2b) [2]	Comments can be combined e.g Both distributions negatively skewed gets both marks e.g. 1974 distribution has greater spread than 2014 gets both marks	If zero scored, SC1 for “The 2014 distribution is shifted to the right of the 1974 distribution” oe	
		b	i Life expectancy went down [between 1974 and 2014] in [at least] one country	E1(AO2.2a) [1]	NOT increase in life expectancy is negative		
		b	ii The box plot is not symmetrical.	B1(AO3.5b) [1]			
		b	iii Not appropriate with reason	E1(AO2.4) [1]	e.g. [some] values of life expectancy are estimates The values of life expectancy are not available to this level of accuracy		
		b	iv Comment about life expectancy at birth data for countries and not individual people	B1(AO2.4) [1]			

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	Use of $Q3 + 1.5 \times (Q3 - Q1)$ $15.873 + 1.5(8.9154) = 29.2461$ (years) The maximum value is an outlier as $30.742 > 29.2461$.	M1(AO1.2) M1(AO1.1) A1(AO1.1) [3]		
		d	i approx $60.8 - 37.5 = 23.3$ (years)	M1(AO3.1b)	Attempt to estimate change in life expectancy at birth soi.	
		d	ii Change in life expectancy for Sweden approx $81.9 - 72.5 = 9.4$ (years)	A1(AO1.1 1.1)	FT 'their 37.5 between 35 - 40'	
		d	iii E.g. Countries with a lower life expectancy in 1974 have greater opportunity to increase life expectancy in 2014.	A1 E1 (AO3.2a) [4]	FT 'their 72.5 between 70 - 75' OR Countries with a higher life expectancy in 1974 have less opportunity to increase life expectancy in 2014.	
		e	i $30.98 + 0.67 \times 37.4 = 56.0$ (years)	M1(AO3.4) A1(AO1.1) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	ii E.g. Large amount of scatter at the lower values [and South Sudan is 37.4]. E.g. Not having the data value could indicate that there are problems in the country which could mean it does not follow the pattern for other countries	E1(AO3.5b) E1(AO3.5b) [2]	E1 Reason inferred from Fig 16.4 E1 For knowing why data may be missing	
		f	Correct method Clearly explained 6	M1(AO3.1b) E1(AO2.4) A1(AO1.1) [3]	e.g. draw “y = x” on graph e.g. The value on the vertical axis must be lower than the one on the horizontal axis FT their correct method	
			Total	20		

Question			Answer/Indicative content	Marks	Part marks and guidance		
28		a		B1(AO1.1a) B1(AO1.1) [2]	Correct shape and symmetry for cosine graph. Correct maximum and minimum values		
		b	DR $2\cos\theta = 3\sin^2\theta$ $2\cos\theta = 3(1 - \cos^2\theta)$ $3\cos^2\theta + 2\cos\theta - 3 = 0$ $\cos\theta = \frac{-1}{3} + \frac{\sqrt{10}}{3}$ $\theta = 43.9^\circ, 316.1^\circ$ $\cos\theta = \frac{-1}{3} - \frac{\sqrt{10}}{3} < -1$ gives no solution	B1(AO1.2) M1(AO3.1a) M1(AO1.1) A1(AO1.1) A1(AO1.1) E1(AO2.4) [6]	Correct use of identity must be seen Rearranging to zero must be seen , condone one error Solve quadratic Or state that graph in part (a) only shows two solutions		

Question			Answer/Indicative content	Marks	Part marks and guidance		
			Total	8			
29			DR $[y = k](x+1)^2(x-2)$ Substitute (0, -3) or (1, -6) $[y = \frac{3}{2}(x+1)^2(x-2)]$ $[y = \frac{3}{2}x^3 - \frac{9}{2}x - 3]$ gradient of tangent is $\frac{dy}{dx} = \frac{3}{2}(3x^2 - 3)$ $f' = (-f)^2$ therefore the gradients are equal and the tangents are parallel	M1*(AO3 .1a) M1*(AO3 .1a) A1dep(AO1.1) M1*(AO1 .1) A1dep(AO2.1) E1(AO2.2a) [6]	FT their f(x) even if the gradient property does not hold for it Not just "the gradient is the same for $-f'$". Allow FT from their f(x) if the gradient property holds		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
30		a	P is $(\sqrt{2}, \frac{\sqrt{2}}{2})$ $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ $= \frac{\cos \theta}{-2 \sin \theta}$ When $\theta = \frac{\pi}{4}$, $\frac{dy}{dx} = -\frac{1}{2}$ Equation of tangent is $(y - \frac{\sqrt{2}}{2}) = -\frac{1}{2}(x - \sqrt{2})$ $\Rightarrow y = -\frac{1}{2}x + \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}$ $\Rightarrow x + 2y = 2\sqrt{2}$	B1(AO1.1) M1(AO3.1a) A1(AO1.1) A1(AO1.1) B1(AO2.1) E1(AO1.1) [6]	oe		
		b	When $x = 0$, $y = \sqrt{2}$ so A is $(0, \sqrt{2})$ When $y = 0$, $x = 2\sqrt{2}$ so B is $(2\sqrt{2}, 0)$ Area of triangle $= \frac{1}{2}\sqrt{2} \times 2\sqrt{2} = 2 \text{ units}^2$	B1(AO1.1) B1(AO1.1) B1(AO1.1) [3]			
			Total	9			

Mark Scheme

Question		Answer/Indicative content	Marks	Part marks and guidance		
31		DR Radius = $\sqrt{10}$ $\cos C = \frac{10+10-(7-1)^2}{2 \times \sqrt{10} \times \sqrt{10}}$ $C = 2.50$ (3sf) Area = $\frac{1}{2} \times (\sqrt{10})^2 \times 2.50 - \frac{1}{2} \times (\sqrt{10})^2 \times \sin 2.50$ Area = 9.49	B1(AO1.1) M1(AO3.1a) A1(AO1.1) M1(AO3.1a) A1(AO1.1) [5]	Or use right angled triangle: $\cos x = \frac{3}{\sqrt{10}}$ and for $\frac{1}{2}C = \frac{\pi}{2} - \cos^{-1}\left(\frac{3}{\sqrt{10}}\right)$		
		Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
32		a	Distance $CO = \sqrt{(-4)^2 + 3^2} = 5$ $(x + 4)^2 + (y - 3)^2 = 25$	M1(AO 1.1a) M1(AO 1.1a) A1(AO 2.1) [3]	Allow $(-4)^2 + 3^2$ if CO^2 clearly stated Use of circle equation with their r^2 Correct answer; isw		
		b	A is the point (0, 6) Gradient of AC is $\frac{3}{4}$ Gradient of tangent is $-\frac{4}{3}$ [Equation of tangent is $y - 6 = -\frac{4}{3}x$] When $y = 0$, $x = \frac{9}{2}$	B1(AO 3.1a) M1(AO 1.1a) M1(AO 1.1a) M1FT(AO 1.1b) A1(AO 2.2a) [5]	Use their point A and gradient to find B cao		
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance	
33			$OP = OQ = 6$ Centre of circle is at (3, 3) $r^2 = 3^2 + 3^2$ $= 18$ $(x - 3)^2 + (y - 3)^2 = 18$	M1(AO 3.1a) M1(AO 3.1a) M1(AO 1.1) A1(AO 1.1) A1(AO 1.1) [5]	Use area of triangle to find OP and OQ Midpoint of PQ oe isw	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance		
34		a	DR At intersections, $x^2 - kx = 3(k + 1) + kx - x^2$ $2x^2 - 2kx - 3(k + 1) = 0$ For touching, '$b^2 - 4ac = 0$' $4k^2 + 24(k + 1) = 0 \Rightarrow k^2 + 6k + 6 = 0$ $k = \frac{-6 \pm \sqrt{12}}{2}$ $k = -3 \pm \sqrt{3}$	M1(AO 3.1a) M1(AO 1.1) M1(AO 2.1) A1(AO 2.2a) B1(AO 1.1) B1(AO 1.1) [6]	Forming quadratic 0 on one side of quadratic Forming quadratic from discriminant Correct quadratic in 3-term form Use of formula or completing the square	$(k + 3)^2 = 3$	
		b	On y-axis $x = 0$, so $y = 0^2 - 0k = 0$ If they cross on the y-axis, they cross at the origin $0 = 3(k + 1) + k \cdot 0 - 0^2$ $k = -1$ [so a value exists]	M1(AO 3.1a) B1(AO 2.2a) M1(AO 1.1) A1(AO 2.1) [4]	Use of $x = 0$ to find y Use of $x = 0$ and $y = 0$		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
35		a	DR $\frac{dy}{dx} = 4x^3 - 6x$ When $x = 1$, $\frac{dy}{dx} = -2$ Gradient of normal is $\frac{1}{2}$ Equation is $y = \frac{1}{2}(x-1)$	M1(AO 1.1) A1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [4]	Attempt to differentiate with at least one term correct Use of m_1 $m_2 = -1$ oe		
		b	DR Curve is part (a) translated up by $\frac{1}{2}$ $c = 2.5$ Alternative method Equation of normal is $y = \frac{1}{2}x$ Point on curve is $(1, \frac{1}{2})$ $c = 2.5$	M1(AO 3.1a) M1(AO 1.1) A1(AO 2.2a) M1 M1 A1 [3]			
			Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance		
36			Use of $-1/$ (their $\frac{1}{2}$) $y - 5 = -2(x - 1)$ $y = -2x + 7$	M1(AO 3.1a) M1(AO 1.1) A1(AO 1.1) [3]			
			Total	3			
37			DR $\frac{dx}{dt} = -\sin t - 3$ $\frac{dy}{dt} = 3 + 4\sin t - 2\cos 2t$ $\frac{dy}{dx} = \frac{3 + 4\sin t - 2\cos 2t}{-\sin t - 3}$ denominator is always negative since $-1 \leq \sin t \leq 1$ Substitution of $\cos 2t = 1 - 2\sin^2 t$ in numerator $4\sin^2 t + 4\sin t + 1$ seen in numerator Numerator is $(2\sin t + 1)^2$ so numerator is always positive and denominator is always negative, so $\frac{dy}{dx}$ is always negative	B1(AO 3.1a) B1(AO 1.1) M1(AO 2.1) E1(AO 2.4) M1(AO 1.1) A1(AO 1.1) E1(AO 2.4) [7]			
			Total	7			

Question		Answer/Indicative content	Marks	Part marks and guidance	
38		correct ruled line of intercept (0, 1) and gradient -2 drawn on grid, extending at least from $(-1.5, 4)$ to $(2, -3)$, as per the circles on the overlay, tol. 1mm horizontally (ie half a square on the grid)	2	M1 for correct line but eg not extending into 4th quadrant or M1 for line with correct gradient but wrong intercept or M1 for line with correct intercept and gradient negative but not -2 or M1 for correct plots but line not ruled	NB page 12 shown in this image must be annotated as BP if blank. Highlight (to indicate seen) in q1 image of page 12 if just rough work crossed out. If a qn number is shown and relevant work seen, highlight the qn no in the q1 image then use full response view to link page 12 to relevant question; also put a highlight by the image in the correct qn space to remind you there is extra work to look at.
			[2]	Examiner's Comments Most candidates coped well with this question, usually scoring full marks. Errors included drawing a line with	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					the wrong gradient, ie +2 or – 1/2 and marks were also dropped due to inaccurately drawn lines based upon plotting one point and roughly estimating where a gradient of –2 would be. It would have been much more accurate for those candidates to plot at least two points when drawing a straight line. If plotting points it would be wise for candidates to only mark a small point, not a large circular blob encompassing a complete square as some candidates did. It was sad to see some candidates not using a ruler and therefore drawing wobbly freehand lines.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
39		i	[centre] $(-2,3)$	B1		
			[radius] $\sqrt{5}$	B1 [2]	B0 for $\pm$ $\sqrt{5}$	
					Examiner's Comments There were a small number of candidates who incorrectly worked out the centre of the circle, usually giving the centre with incorrect signs. A number of candidates gave the radius as either 5 or 25 but most candidates scored full marks here. The vast majority of candidates found the correct equation of the required line, with those who dropped marks usually because they chose 5 as the gradient and not -5 .	
		ii	$5x + y = -7$ or $y = -5x - 7$ or $5x + y + 7 = 0$	2 [2]	M1 for $5x + y = k$, $k \neq 4$ or for gradient of parallel line $= -5$ or for answer $-5x - 7$	if wrong centre in 5(i), can earn just M1
					Examiner's Comments Some of these candidates clearly misunderstood the concept of $y = mx + c$ as they clearly believed that the coefficient of x was the gradient no matter what side of the equation the x term appeared on. Only a small minority used a gradient based on the negative reciprocal.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	4	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
40		i	$AB^2 = 5^2 + 5^2 = 50$ $BC^2 = 7^2 + 1^2 = 50$	B1 B1 [2]	oe with AB; may go straight from correct unsimplified form to 50 with no interim working (applies to both marks), but for 2 marks any interim working must be correct oe with BC	for 2 marks to be awarded, notation used must be fully correct. Penalise only one mark if squares and square roots eg 50 and $\sqrt{50}$ confused, or brackets used incorrectly or AB and BC missing, etc, but working is otherwise correct Examiner's Comments Many obtained two marks here without any difficulty. Candidates who used less formal notation often lost marks due to missing brackets or confusion about whether they were working with AB or AB^2. A few candidates confused lengths and gradients.

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	grad AC = $\frac{-1-3}{5-3} [= -2]$ oe isw grad perp = $\frac{1}{2}$ or ft from their grad AC or finding gradient of their BF $y + 2 = \text{their } \frac{1}{2}(x + 2)$ or $-2 = \text{their } \frac{1}{2}(-2) + c$ oe $y = \frac{1}{2}x - 1$ isw	B1 M1 M1 A1 [4]	or midpt F of $AC = \left(\frac{5+3}{2}, \frac{3-1}{2} \right)$ may be seen in eqn of perp or using coords of their F; no ft for using their grad AC for this allow both M1s for eqn of line through B and their F Examiner's Comments This was completed well by the majority of candidates. A few quoted the gradient formula incorrectly or had difficulty simplifying the gradient accurately, but were then able to find the associated perpendicular gradient and use the equation of a straight line well.	must be a changed gradient related to grad AC, or be their grad BF, to score this M1	
		iii	(10, 4)	2	B1 for each coordinate or M1 for use of $\overline{AD} = \overline{BC}, \overline{CD} = \overline{BA}$ or $\overline{BF} = \overline{FD}$ or for correct method for intersection of (ii) line		

Question			Answer/Indicative content	Marks	Part marks and guidance		
					and line through A parallel to BC $[y = 3 + \frac{1}{7}(x - 3)]$ oe or $y = \frac{1}{7}x + \frac{18}{7}$ oe if correct] or line through C parallel to BA $[y = x - 6]$ if correct] allow SC1 for $(-4, 2)$ for ADBC found, or $(0, -6)$ for ABDC found [both parallelograms, not rhombi] Examiner's Comments This presented a challenge to a significant number of candidates, with those who chose not to use a vector related method often getting bogged down with complicated algebra. A common error was to not appreciate the importance of the letter order ABCD, and instead give ACDB or ACBD, which earned partial credit but affected the difficulty of part (iv), so limiting the marks available there.	NB more complicated methods exist using simultaneous equations and eg $\text{grad BD} = \frac{1}{2}$ and $AD^2 = BC^2$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		iv	grad $AD = \frac{4-3}{10-3}$ or $\frac{1}{7}$ or ft relevant D from attempt at ABCD	M1	however, if D = (0, -6), or (-4, 2) or other attempt at ABDC or ADBC in (iii), or other attempt where one or both coords of D are less than the co respondin g coords of (8, 3.8) award only SC1 in (iv) for showing by diagram or coordinates that E is obviously outside the rhombus ABDC eg since its x -coordinate is greater than the x -coordinate of all the vertices (or similarly y -coordinate s)	some are working with CD only, not AD. Give M0 but allow SC1 for showing that CD is y = $x - 6$ and then finding on CD when $y =$ 3.8, $x = 9.8$ or when $x =$ 8, $y = 2$; allow ft from wrong but relevant D – see 'however' in previous column	
			so when $x = 8$, y-coord. on AD = $3 + \frac{1}{7} \times (8 - 3)$ or ft	M1	or use of $y - 3 = \frac{1}{7}(8 - 3)$ oe	may use coords of their D not A in eqn	
				A1	or M1 for	i.e. M1 for substituting one coord of E in their equation for AD after correct	

M1	8 n w n w
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Question			Answer/Indicative content	Marks	Part marks and guidance		
			grad. AE shown to be greater than grad AD eg $0.16 > 0.14...$ or grad DE shown to be less than grad AD eg $0.1 < 0.014...$ conclusion E is outside rhombus	A1 [4]		$3.8 - \frac{8}{7} > \frac{18}{7}$, A1 for conclusion E is outside rhombus the SC for working with CD rather than AD is also available if they use gradients – allow SC1 if they find grad CD = 1, and grad CE = $4.8/3$ or 1.6 or grad DE $\frac{0.2}{2} [= 0.1]$ no ft from wrong D; no ft from wrong D Examiner's Comments This part required candidates to apply some reason and insight rather than just applying well-drilled techniques. Candidates would have found it helpful to sketch a diagram with their D marked, to ensure that they were comparing E to the correct line (AD). Most started by finding the equation of AD and a good number successfully used this to decide whether E was above or below AD.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Some who substituted $x = 8$ into AD found $y = 3\frac{5}{7}$ but did not prove that this is less than 3.8. Some compared with CD rather than AD. Some candidates used other methods, often efficiently, such as showing that the gradient of AE was greater than the gradient of AD.	
			Total	12		
41		a	$(x - 2)^2 + (y + 3)^2$ seen (2, -3)	M1(AO1.1) A1(AO1.1b) [2]	ignore other working	
		b	$(x - 2)^2 + (y + 3)^2 - 2^2 - 3^2 - 12 = 0$ or better seen $r = 5$	M1(AO1.1b) A1 (AO1.1b) [2]		
			Total	2		